# Asymmetric Capital Buffer Control

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## 1 Introduction

### 1.1 Question

When a bank's capital buffer is explained by a fixed issuance cost together with a shortfall penalty that grows as equity falls below required capital, does the bank's policy carry the asymmetry parameter of that penalty separately from the fixed issuance cost, and if so where, in the dividend barrier and the recapitalisation trigger or in buffer variance? The asymmetry parameter is never observed (Appendix C), so it can be read only off what the policy shows, and at one it switches the shortfall penalty off and leaves the model with issuance costs alone (M8), so whether it is identified decides whether the specification Peura and Keppo [2006] call for (L18) can be tested against theirs rather than imposed (L20). The bank recapitalises by impulse control at a fixed issuance cost and takes risky exposure under a regulatory cap, and the cap sets the multiple of asset volatility that its equity bears, the mechanical leverage that Engle and Siriwardane [2018] separate from exposure to the market in the asymmetry of equity volatility (L13, L34).

### 1.2 Answer

The shortfall penalty moves the thresholds, the dividend barrier and the recapitalisation trigger. The asymmetry parameter $\lambda_S$ and the fixed issuance cost $K$ both raise the dividend barrier (M1). The recapitalisation trigger rises in $\lambda_S$ and falls in $K$ (M2). Because the two parameters move the trigger in opposite directions, the Jacobian of the barrier and the trigger in $(\lambda_S, K)$ is non-singular, so the pair is locally identified from the two thresholds (M3), under the hypotheses that M4 and Appendix B list. This is K1 and C1. The thresholds identify the asymmetry parameter and not whose cost the shortfall penalty is (Appendix D). The identification is within the model. The two thresholds are recorded as the state at a dividend and at an issue, and on the FDIC panel of US banks of 2015 to 2024 dividends are paid at every state and issues land at no single target, so neither is measured there as a threshold (Appendix C).

The shortfall penalty does not move the limits of buffer variance. Where the optimal exposure binds the cap near the distress boundary and deep in surplus, buffer variance tends to the deficit limit and the surplus limit, which involve only the asset volatility, the liability volatility, the return-liability correlation and the regulatory cap (M10, M11). For every exposure that complies with the cap, binding or not, buffer variance is bounded by the same quantities (M12). The identities hold at each date, so with an asset volatility that varies over time buffer variance is the limit at that date's asset volatility wherever the cap binds (M15). The asymmetry parameter enters the thresholds and not the limits of buffer variance. This is K2 and C2, with its scope in C3. The limits need the cap to bind, and on that panel the recorded exposure sits far below the cap, so the bound of M12, which needs no binding, is the result that reaches the data (Appendix C).

Table 1: Headlines.

| ID | Headline | Model rows | Scope note |
|---|---|---|---|
| K1 | The asymmetry parameter $\lambda_S$ and the fixed issuance cost $K$ are locally identified from the dividend barrier and the recapitalisation trigger, because both raise the barrier and they move the trigger in opposite directions. | M1, M2, M3, M4 | Local identification from the thresholds, not an estimator. Rests on the hypotheses of Appendix B, among them the verification theorem, the envelope identities, the signs of $W'(x_L)$ and $N'$, $\Psi(x_L) > 0$ and $\rho_L > \mu'(y^*)$. Fails if the two parameters moved the trigger in the same direction (M3). It identifies them given the asset volatility, which moves the two thresholds as the fixed issuance cost does, and given no delay on issues, which the thresholds read as a higher asymmetry parameter (Section 3). It identifies the asymmetry parameter, not whose cost the shortfall penalty is (L19, Appendix D). The thresholds are recorded as the state at a dividend and at an issue, and on the FDIC panel of US banks of 2015 to 2024 neither behaves as a threshold (Appendix C), so the identification is within the model. It needs the recapitalisation trigger above the distress boundary. In the solve of Appendix B the trigger is interior at 42 of 56 parameter points, and where the fixed issuance cost is large against the asymmetry parameter the bank issues only at the distress boundary, where the trigger cannot move. The fixed issuance cost is charged per unit of liabilities (Appendix D). The responses of the thresholds to the fixed issuance cost alone are not claimed new, since Peura and Keppo [2006] and Dai et al. [2019] tabulate them, nor is pinning the fixed issuance cost and a second parameter from moments of the thresholds, which Peura and Keppo [2006] do by calibration (L7, L8, L9). |
| K2 | The asymmetry parameter enters the thresholds and not the limits of buffer variance. Where the optimal exposure binds the regulatory cap near both ends, those limits depend only on the asset volatility, the liability volatility, the return-liability correlation and the regulatory cap, and for every compliant exposure buffer variance is bounded by them. | M3, M10, M11, M12, M15 | That the optimum binds the cap near both ends is a hypothesis. The limits are instantaneous, so with an asset volatility that varies over time they hold at each date at that date's asset volatility wherever the cap binds (M15); that the optimum binds the cap when the asset volatility moves is not shown. In the solve of Appendix B the exposure is at the cap at both thresholds and on 85 percent of the inaction region; the bound M12 needs no binding. The surplus limit needs the liquidity branch to bind deep in surplus, which needs borrowing; Bayraktar et al. [2026] take no borrowing as the main banking case, under which it never binds (L2, Appendix D). The constant buffer variance on the solvency branch is not claimed new, since it is the constant log-cushion volatility of capped portfolio insurance [Black and Perold, 1992], L5. The contrast is with that rule, whose slopes are set by preferences (L6), while here they are set by regulation and the liability shock is not hedged. On the FDIC panel of US banks of 2015 to 2024 the cap does not bind, so only the bound of M12 reaches the data (Appendix C). |

# 2 Model

## 2.1 Players

Table 2: Players of the model, the choice each faces and how the rows settle it.

| ID | Player | Choice faced | How the model settles it | In the literature | Rows |
|---|---|---|---|---|---|
| P1 | Bank | Its risky exposure, the dividend barrier, the recapitalisation trigger and the injection target. | The thresholds are derived, in their responses to $\lambda_S$ and $K$ (M1 to M4) and in the target's independence of the starting state (M7). The exposure is not derived. M11 takes it at the cap as a hypothesis, and M12 bounds every compliant exposure. | The bank of Peura and Keppo [2006] and of Dai et al. [2019], with a fixed issuance cost, and of Bayraktar et al. [2026], with the cap. | M1, M2, M3, M4, M7, M11, M12, L1, L7, L9 |
| P2 | Shareholders | None in the model. They receive dividends and supply each issue at the fixed and the proportional issuance cost. | A rule taken as given. The objective is their value net of the shortfall penalty, and the bank chooses for them (M8). Finite issuance follows from the fixed issuance cost (M6). | In Peura and Keppo [2006] the owners choose the issue. | M6, M8, L7 |
| P3 | Creditors | None. The liabilities carry their own shock. | State. The liability volatility and the return-liability correlation enter the limits and bounds of buffer variance (M10 to M12). | No paper read has a liability shock unhedged by the safe asset; in Black and Perold [1992] the reserve asset tracks the liability. | M10, M11, M12, L5 |
| P4 | Supervisor | None in the model. It sets required capital and the regulatory cap. | A rule taken as given (A2). The shortfall penalty below the reference stands for its restrictions, a reading no row derives (Appendix D). | The cap of Bayraktar et al. [2026]. The intervention of Milne and Whalley [2001], which the buffer guards against (L16). | M8, M10, L1, L16 |

Figure 1 draws the model as a graph. Its nodes are the agents and the object they act on, equity, named as in Appendix C, and its arrows are the actions between them, each labelled by the rows that settle it. The players of Table 2 are its agents, with the creditors drawn as the insured and the uninsured, since they hold the liabilities differently and the model takes their shock as one block (A1).

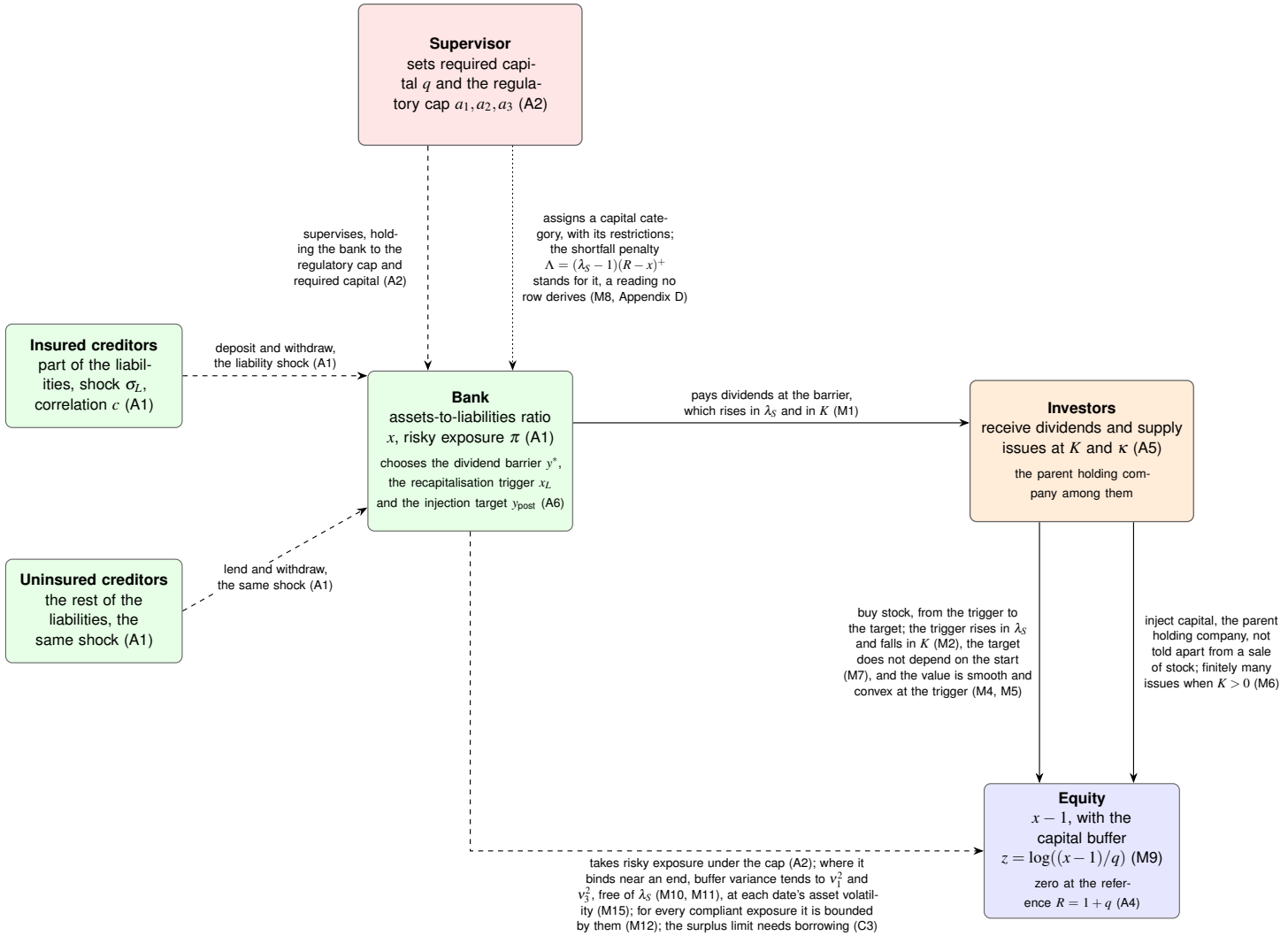



Figure 1: The paper's ledger. Six entities, the agents in green and orange, the supervisor in red and equity in blue, and eight actions. Solid arrows are settled by model rows, dashed arrows are rules or inputs taken as given, and the dotted arrow is open. Each label cites the rows that settle the action. The insurer is not drawn, since no row bears on it. K1 concerns the dividend arrow and the two issue arrows together (M3), and K2 the exposure arrow and equity (M10 to M12).

## 2.2 Assumptions

Table 3: Assumptions of the model.

| ID | Assumption | In plain English | Symbols | Model rows | Reason |
|---|---|---|---|---|---|
| A1 | The state is $x \in (1,\infty)$, with drift $\mu(x,\pi) = x((1-\pi)r+\pi\mu_s-\mu_L)+\gamma$ and diffusion $\sigma^2(x,\pi) = \pi^2\sigma^2x^2+2\pi c\sigma\sigma_L x(1-x)+\sigma_L^2(1-x)^2$ under risky exposure $\pi$, and the capital buffer is $z=\log((x-1)/q)$ with $q>0$. | The bank holds a risky and a riskless asset and owes liabilities that carry their own shock. The state is the assets-to-liabilities ratio. The riskless rate, the expected asset return, the liability drift and the net income on liabilities set the drift of the state, and the risky exposure sets how much of the asset volatility the bank bears beside the liability volatility and the return-liability correlation, and the capital buffer measures equity against required capital on a log scale. | $x$, $\pi$, $\mu$, $r$, $\mu_s$, $\mu_L$, $\gamma$, $\sigma$, $\sigma_L$, $c$, $q$, $z$ | M1, M4, M9, M10, M11, M12, M13, M14, M15 | The balance sheet of Bayraktar et al. [2026] in ratio form, their Proposition 2.1. Appendix D, the balance sheet. |
| A2 | The regulatory cap is $u(x)=\min\{(x-1)/a_1,(x-a_2)/a_3\}$ with $0<a_1<a_3$ and $a_2<1$, and the exposure complies, $0\le\pi(x)x\le u(x)$. | Regulation sets a regulatory cap on the risky exposure at each assets-to-liabilities ratio, the smaller of a solvency branch, whose slope is the solvency floor and which binds near distress, and a liquidity branch, set by the run-off rate and the liquidity haircut and binding deep in surplus. | $u$, $a_1$, $a_2$, $a_3$, $\pi$, $x$ | M10, M11, M12, M13, M15 | The cap of Bayraktar et al. [2026]. Appendix D, the regulatory cap. |
| A3 | Buffer variance is $\zeta(x)=z'(x)^2\sigma^2(x,\pi(x))$. | Buffer variance is the variance rate of the capital buffer, by the change of variable from the assets-to-liabilities ratio, at the risky exposure the bank takes and the asset volatility it bears. | $\zeta$, $z$, $x$, $\pi$, $\sigma$ | M10, M11, M12, M13, M14, M15 | Itô's formula, not proved in Lean. Appendix B. |
| A4 | The objective is dividends net of issuance costs minus the running shortfall penalty $\Lambda(x)=(\lambda_S-1)(R-x)^+$ with $\lambda_S\ge 1$ and reference $R=1+q$, discounted at $\rho_L$. | Shareholders value dividends net of what they pay for issues, less a shortfall penalty for each moment the assets-to-liabilities ratio sits below the reference, the state at which equity equals required capital, weighted by the asymmetry parameter and discounted at the liability-net discount rate. | $\Lambda$, $\lambda_S$, $R$, $q$, $\rho_L$, $x$ | M1, M2, M3, M4, M8 | The asymmetric penalty is the paper's question. Appendix D, the shortfall penalty. |
| A5 | Each issue costs $K$ per unit of liabilities plus $\kappa$ per unit raised. | Every recapitalisation pays a fixed issuance cost that does not scale with its size and a proportional issuance cost. | $K$, $\kappa$ | M2, M5, M6, M7 | The cost structure of Peura and Keppo [2006] and Bolton et al. [2011]. Appendix D, the fixed issuance cost. |

| ID | Assumption | In plain English | Symbols | Model rows | Reason |
|---|---|---|---|---|---|
| A6 | $V$ is the value function, solves the quasi-variational inequality with impulse operator $\mathscr{M}V$, is twice continuously differentiable on the inaction region, and the optimal policy is a dividend barrier $y^*$, a trigger $x_L$ and a target $y_{\text{post}}$, with $\partial_{\lambda_S} V = -W$ and $\partial_K V = -N$. | The value function is the value of the bank under its best policy, which pays dividends at the dividend barrier and recapitalises from the recapitalisation trigger to the injection target, the impulse operator being the value of issuing at once, and the value responds to the asymmetry parameter and the fixed issuance cost through the expected discounted shortfall and the discounted injection count. | $V$, $\mathscr{M}V$, $y^*$, $x_L$, $y_{\text{post}}$, $W$, $N$, $\lambda_S$, $K$ | M1, M2, M3, M4, M5 | The verification theorem and the envelope identities, not proved in Lean. Appendix B. |

## 2.3 The bank's problem

One player optimises. The bank chooses its risky exposure $\pi$, its dividends $D$ and the times $\tau_n$ and sizes $I_n$ of its recapitalisations, and it chooses them for its shareholders, who receive the dividends and pay for each issue (P1, P2). The supervisor sets required capital and the regulatory cap before the problem starts and does not act within it (P4, A2), and the creditors choose nothing (P3). The bank also chooses when to recapitalise, where in Bayraktar et al. [2026] the supervisor fixes the state at which it must (L28).

Between issues the assets-to-liabilities ratio follows

$$dx_t = \mu(x_t,\pi_t)\,dt + \pi_t x_t \sigma\,dB_t + \sigma_L(1-x_t)\,dW_t - dD_t, \qquad \mu(x,\pi) = x\big((1-\pi)r + \pi\mu_s - \mu_L\big) + \gamma,$$

with $d[B,W]_t = c\,dt$, and the issue at $\tau_n$ moves it from $x_{\tau_n-}$ to $x_{\tau_n-} + I_n$. The quadratic variation of $x$ per unit of time is the diffusion $\sigma^2(x,\pi)$ of A1, and $\mu(x,\pi)$ is the drift of the state, which along the cap is the drift $\mu$ of Appendix A. This is the reduction of Bayraktar et al. [2026] to one state, their Proposition 2.1, in which values are per unit of liabilities and expectations are under the measure that weights each path by its liabilities. Their dilution cost is replaced by the fixed and the proportional issuance costs of A5, and the fixed issuance cost is charged per unit of liabilities so that the reduction survives it (L4, Appendix D).

A policy is admissible when the exposure complies with the regulatory cap at every date, $0 \le \pi_t x_t \le u(x_t)$ (A2), the dividends are non-decreasing and right-continuous with $D_{0-} = 0$, the issue times are increasing stopping times and each issue size is known at its issue time. The bank maximises

$$V(x) = \sup\ \mathbb{E}_x\Big[\int_0^\infty e^{-\rho_L t}\big(dD_t - \Lambda(x_t)\,dt\big) - \sum_{n\ge 1} e^{-\rho_L \tau_n}\big(K + (1+\kappa)I_n\big)\Big]$$

over admissible policies. The integral is the dividends less the shortfall penalty of A4, and the sum is what the shareholders pay for each issue (A5). The liability-net discount rate is $\rho_L = \rho - \mu_L$, the shareholders' discount rate net of the liability drift, since values are per unit of liabilities.

The shortfall penalty is part of what the bank maximises and not a payment anyone receives. At $\lambda_S = 1$ it vanishes and the objective is the shareholders' value alone (M8). Above one it charges each moment the state spends below the reference, and the model does not say whose cost that charge stands for, the supervisor's restrictions below required capital or the market's response. Appendix D adopts the first reading without deriving it, and K1 identifies the asymmetry parameter under any of them. Appendix D also says why the objective takes this form.

At the distress boundary the regulatory cap is zero, so the exposure is zero, the diffusion vanishes and the drift is $r - \mu_L + \gamma$. When that drift is positive, as at every parameter point of Appendix B, the boundary repels the state locally [Bayraktar et al., 2026, Remark 3.1]. The problem has no liquidation, where Peura and Keppo [2006] liquidate at the minimum (L24) and in Fries et al. [1997] the shareholders stop injecting capital and close the bank at a point they choose (L75).

The dynamic programming equation of the problem is the quasi-variational inequality

$$\max\Big\{\sup_{0\le\pi x\le u(x)}\big[\tfrac{1}{2}\sigma^2(x,\pi)V'' + \mu(x,\pi)V'\big] - \rho_L V - \Lambda(x),\ \ \mathscr{M}V - V,\ \ 1 - V'\Big\} = 0, \tag{1}$$

with $\mathscr{M}V(x) = \sup_{\xi>0} V(x+\xi) - K - (1+\kappa)\xi$ the impulse operator, the value of issuing at once. Its first term is waiting with the best compliant exposure, its second is issuing at once and its third is paying a dividend. A6 takes

as given that $V$ solves it and that the optimal policy is a dividend barrier $y^*$, a recapitalisation trigger $x_L$ and an injection target $y_{\text{post}}$. Under that policy the state stays in $[x_L, y^*]$, a dividend is paid whenever the state reaches $y^*$, and when it falls to $x_L$ the bank issues $y_{\text{post}} - x_L$. The exposure is the maximiser in the first term, which no row derives (P1).

Rows M1 to M7 are statements about the thresholds of this policy, and M10 to M12 and M15 about buffer variance under the exposure. Section 3 solves equation (1) numerically and varies the asymmetry parameter, the fixed issuance cost, the cap geometry and required capital in it.

## 2.4 Results

Table 4: Results of the model, each with its statement and the Lean theorems that prove it.

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M1 | The dividend barrier rises in the asymmetry parameter and in the fixed issuance cost. | $\theta \in \{\lambda_S, K\}$, $F = W$ or $N$. If $V''(y^*) = 0$, $V'(y^*) = 1$, $\Lambda'(y^*) = 0$, $F'(y^*) = 0$, $\sigma^2(y^*) > 0$, $\rho_L \neq \mu'(y^*)$, $\partial_\theta V = -F$ and the chain rule holds along $y^*(\theta)$, then $dy^*/d\theta = \rho_L F(y^*)/(\rho_L - \mu'(y^*))$, positive when $0 < \rho_L$, $\mu'(y^*) < \rho_L$ and $F(y^*) > 0$. | `Identification.vppp_at_barrier`, `Identification.fpp_at_barrier`, `Identification.barrier_slope`, `Identification.barrier_slope_pos`, `Identification.control_barrier_sign_needs_gap`, `Envelope.wode_of_envelope`, `Envelope.node_of_envelope`, `Envelope.neumann_at_ystar` | p. 46 |
| M2 | The recapitalisation trigger rises in the asymmetry parameter and falls in the fixed issuance cost, the injection target rises in the fixed issuance cost, and the trigger-to-target gap widens in it. | If $V''(x)\,dx/d\theta + \partial_\theta V'(x) = 0$ along a boundary $x(\theta)$, $V''(x) \neq 0$ and $\partial_\theta V' = -F'$, then $dx/d\theta = F'(x)/V''(x)$. With $W'(x_L) > 0$, $N'(x_L), N'(y_{\text{post}}) < 0$ and $V''(x_L) > 0 > V''(y_{\text{post}})$, $dx_L/d\lambda_S > 0$, $dx_L/dK < 0$, $dy_{\text{post}}/dK > 0$ and $d(y_{\text{post}} - x_L)/dK > 0$. | `Identification.boundary_slope`, `Identification.trigger_rises_in_lambda`, `Identification.trigger_falls_in_K`, `Identification.target_rises_in_K`, `Identification.gap_increasing_in_K`, `Identification.control_trigger_slope_needs_curvature`, `Envelope.trigger_jump` | p. 46 |
| M3 | The Jacobian of the barrier and the trigger in the two parameters is non-singular, because the parameters move the trigger in opposite directions. | $\det \partial(y^*, x_L)/\partial(\lambda_S, K) = \frac{\rho_L}{(\rho_L - \mu'(y^*))V''(x_L)}\big(W(y^*)N'(x_L) - N(y^*)W'(x_L)\big)$, negative under the signs of M1 and M2. For $a_{11}, a_{12}, a_{21} > 0 > a_{22}$, $a_{11}a_{22} - a_{12}a_{21} < 0$. With $a_{22} > 0$ it can vanish. | `Identification.det_formula`, `Identification.det_neg_of_opposite`, `Identification.det_neg`, `Identification.control_same_direction_can_fail` | p. 47 |
| M4 | The value function is convex just above the recapitalisation trigger, strictly so exactly when waiting is strictly suboptimal just below it. | $\Psi = \rho_L V + \Lambda - (1+\kappa)\mu$. If $\sigma^2(x_L) > 0$, $\frac{1}{2}\sigma^2 V'' + \mu V' - \rho_L V - \Lambda = 0$ at $x_L^+$, $V'(x_L) = 1 + \kappa$ and $\mu(1+\kappa) - \rho_L V - \Lambda \leq 0$ at $x_L$, then $\frac{1}{2}\sigma^2 V''(x_L^+) = \Psi(x_L) \geq 0$, and $V''(x_L^+) > 0 \iff \Psi(x_L) > 0$. With $\Psi(x_L) = 0$, $V''(x_L^+) = 0$. | `SecondOrder.psi_identity`, `SecondOrder.psi_nonneg_of_qvi`, `SecondOrder.trigger_convex`, `SecondOrder.trigger_strict_iff`, `SecondOrder.control_trigger_flat_when_psi_zero` | p. 47 |

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M5 | The value function has no kink at the recapitalisation trigger. | If $V \geq \mathscr{M}V$, $V(x_L) = \mathscr{M}V(x_L)$, $\mathscr{M}V$ is affine with slope $1+\kappa$, $V$ has right derivative $V'(x_L^+)$, $\sigma^2(x_L) > 0$, and $V'(x_L^+) > 1+\kappa$ would make $\frac{1}{2}\sigma^2 q + (1+\kappa)\mu - \rho_L V - \Lambda \leq 0$ hold at $x_L$ for every $q$, then $V'(x_L^+) = 1+\kappa$. If $\frac{1}{2}\sigma^2 q + (1+\kappa)\mu - \rho_L V - \Lambda \leq 0$ is needed only for $q \leq 0$, it holds when $(1+\kappa)\mu \leq \rho_L V + \Lambda$. Without $V \geq \mathscr{M}V$ a downward kink is possible. | `SmoothFit.hasDeriv_Mv`, `SmoothFit.kink_is_convex`, `SmoothFit.no_convex_kink`, `SmoothFit.smooth_fit`, `SmoothFit.control_kink_needs_obstacle`, `Envelope.smf_permits` | p. 47 |
| M6 | With a positive fixed issuance cost and a finite value, the discounted number of recapitalisations is finite, so the impulse terms of the verification argument converge absolutely. | $K > 0$, $e_i \geq 0$, $K\sum_{i<n} e_i \leq B$ for all $n \Rightarrow$ $\sum_{i<n} e_i \leq B/K$, $N = \sum_i e_i$ converges and $N \leq B/K$. If also $\lvert c_i \rvert \leq C e_i$, $\sum_i c_i$ converges absolutely. With $K = 0$, $e_i = 1$ the cost bound holds and $N$ diverges. | `ImpulseCount.partial_sums_bounded`, `ImpulseCount.count_summable`, `ImpulseCount.count_le`, `ImpulseCount.impulse_sum_summable`, `ImpulseCount.control_count_needs_fixed_cost` | p. 47 |
| M7 | With a fixed issuance cost and a linear proportional issuance cost, the injection target does not depend on the state from which the bank injects. | $\forall z\ \big(V(z) - (1+\kappa)z \leq V(y) - (1+\kappa)y\big) \iff$ $\forall z\ \big(V(z) - (1+\kappa)(z-x) - K \leq$ $V(y) - (1+\kappa)(y-x) - K\big)$, for every $x$. A target optimal from one $x$ is optimal from every $x'$. With the convex cost $(z-x)^2$ and $V(z) = 2z$, the optimal target moves with $x$. | `Targeting.target_free_of_start`, `Targeting.target_same_for_all_starts`, `Targeting.control_target_moves_with_convex_cost` | p. 48 |
| M8 | At $\lambda_S = 1$ the shortfall penalty vanishes identically, and for every $\lambda_S \geq 1$ the running payoff is concave. | $\Lambda(x) = (\lambda_S - 1)(R-x)^+ \Rightarrow \Lambda \equiv 0$ at $\lambda_S = 1$, and $-\Lambda$ is concave on $\mathbb{R}$ for $\lambda_S \geq 1$. At $\lambda_S = 0$, $R = 0$ it is not concave. | `QVI_Part1.lambda_asymmetry_vanishes_at_one`, `QVI_Part1.neg_Lambda_concave`, `QVI_Part1.control_concavity_needs_lambda_ge_one` | p. 48 |
| M9 | The capital buffer maps the state space onto the real line, with required capital at zero and the distress boundary at minus infinity. | $q > 0 \Rightarrow z$ is strictly increasing on $(1,\infty)$, $1 + qe^{z(x)} = x$ for $x > 1$, $z(1+qe^w) = w$, $z(1+q) = 0$, $z(x) \to -\infty$ as $x \to 1^+$ and $z(x) \to \infty$ as $x \to \infty$. | `CapGeometry.zmap_inverse`, `CapGeometry.zmap_of_inverse`, `Coordinate.zmap_strictMonoOn`, `Coordinate.zmap_at_requirement`, `Coordinate.zmap_tendsto_atBot`, `Coordinate.zmap_tendsto_atTop` | p. 48 |
| M10 | The solvency branch of the cap binds exactly up to the crossing point $\bar{x}$, and there buffer variance is the constant deficit limit $\nu_1^2$. | $0 < a_1 < a_3$, $a_2 < 1 \Rightarrow$ $\bar{x} - 1 = a_1(1-a_2)/(a_3 - a_1) > 0$, $u(x) = (x-1)/a_1$ for $x \leq \bar{x}$, $u(x) = (x-a_2)/a_3$ for $x \geq \bar{x}$. If also $q > 0$, $1 < x \leq \bar{x}$ and $\pi(x)\,x = u(x)$, then $\zeta(x) = \nu_1^2$. | `CapGeometry.hasDerivAt_zmap`, `CapGeometry.zvol2_eq`, `CapGeometry.xbar_sub_one`, `CapGeometry.one_lt_xbar`, `CapGeometry.solvency_binds`, `CapGeometry.liquidity_binds`, `CapGeometry.zvol2_solvency` | p. 48 |
| M11 | For any exposure policy that binds the cap near an end of the state space, buffer variance tends to the deficit limit $\nu_1^2$ at the distress boundary and to the surplus limit $\nu_3^2$ deep in surplus. | $q > 0$, $0 < a_1 < a_3$, $a_2 < 1$. If $\pi(x)\,x = u(x)$ for all $x$ near $1^+$ then $\zeta(x) \to \nu_1^2$ as $x \to 1^+$. If $\pi(x)\,x = u(x)$ for all large $x$ then $\zeta(x) \to \nu_3^2$ as $x \to \infty$. Both fail without $a_1 < a_3$ or without binding. | `CapGeometry.deficit_limit`, `CapGeometry.surplus_limit`, `CapGeometry.variance_ratio_is_cap_geometry`, `CapGeometry.control_surplus_limit_needs_a1_lt_a3`, `CapGeometry.control_constancy_needs_binding` | p. 49 |

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M12 | For every exposure policy that complies with the cap, binding or not, buffer variance is at most the larger of $\sigma_L^2$ and the deficit limit $v_1^2$ everywhere, and deep in surplus at most the larger of $\sigma_L^2$ and the surplus limit $v_3^2$ up to any margin. | $q>0, a_1>0, x>1, 0\le\pi(x), \pi(x)x\le u(x) \Rightarrow \zeta(x)\le\max\{\sigma_L^2, v_1^2\}$. If $a_3>0$, $a_2<1$ and compliance holds for all large $x$, then for every $\varepsilon>0$, $\zeta(x)\le\max\{\sigma_L^2, v_3^2\}+\varepsilon$ for all large $x$. Both fail with $\pi<0$ or with $\pi x>u(x)$. | `Bounds.qv_le_max`, `Bounds.zvol2_eq_qv`, `Bounds.nu2_eq_qv`, `Bounds.qv_zero`, `Bounds.deficit_bound`, `Bounds.surplus_bound`, `Bounds.control_bound_needs_long_exposure`, `Bounds.control_bound_needs_compliance` | p. 49 |
| M13 | At the baseline of Bayraktar et al. the deficit limit is about 3.140 and the surplus limit about 0.0688, a ratio of about 45.6. | $(\sigma,\sigma_L,c,a_1,a_3)=(0.08,0.03,0.20,0.045,0.30)$ $\Rightarrow v_1^2=2543449/810000$, $v_3^2=6193/90000$, $v_1^2/v_3^2=2543449/55737$. | `Comparison.nu1_baseline`, `Comparison.nu3_baseline`, `Comparison.ratio_baseline` | p. 49 |
| M14 | With no liability shock the limits of buffer variance are the log-cushion variances of capped portfolio insurance with multiples $1/a_1$ and $1/a_3$, and a liability shock moves them. | $v_i^2\big\vert_{\sigma_L=0}=(1/a_i)^2\sigma^2$, the log-cushion variance $(E/C)^2\sigma^2$ on the multiplier branch with multiple $1/a_i$. At the baseline of M13 these are $256/81$ and $16/225$. At $(\sigma,\sigma_L,c,a)=(1,1,0,1)$, $v^2=2\ne1=v^2\big\vert_{\sigma_L=0}$. | `Comparison.nu2_hedged`, `Comparison.hedged_is_cushion_vol`, `Comparison.hedged_baseline`, `Comparison.control_liability_shock_moves_nu` | p. 50 |
| M15 | Where the exposure binds the cap, buffer variance at each date is the limit at that date's asset volatility, whatever path the asset volatility follows and whatever policy binds. | $q>0, 0<a_1<a_3$. For any path $t\mapsto\sigma_t$ and exposures $\pi_t$, if $1<x_t\le\bar{x}$ and $\pi_t(x_t)x_t=u(x_t)$ then $\zeta_t(x_t)=v_1^2$ evaluated at $\sigma_t$, the same for any two exposures that bind there, and if $\pi_t(x)x=u(x)$ for all large $x$ then $\zeta_t(x)\to v_3^2$ evaluated at $\sigma_t$. In the asset variance $\sigma_t^2$, $v_1^2=\sigma_t^2/a_1^2-2c\sigma_t\sigma_L/a_1+\sigma_L^2$. With $\pi\equiv0$ at $(\sigma,\sigma_L,c,q,a_1,x)=(1,1,0,1,\frac{1}{2},2)$, $\zeta=1\ne5=v_1^2$. | `TimeVarying.deficit_limit_at_date`, `TimeVarying.same_variance_any_binding_policy`, `TimeVarying.surplus_limit_at_date`, `TimeVarying.nu2_of_variance`, `TimeVarying.control_date_identity_needs_binding` | p. 50 |

# 3 Comparative statics

The section asks three questions of the model that the papers read ask of their own. The first is how the dividend barrier, the recapitalisation trigger and the injection target move with each friction, which the models with issuance costs tabulate for their frictions and not for an asymmetry parameter. The second is how the thresholds and the value function move with the cap geometry and with required capital, which is Table 1 of Bayraktar et al. [2026] with an exogenous trigger and no asymmetry parameter. The third is how the value function trades against safety as the asymmetry parameter rises, where their frontier moves the cap geometry instead. It answers first with what the rows prove and then with the model solved at one baseline. Every number comes from a script in `theory/solve/`, and Table 6 lists the script and the saved output behind each figure and table. No row rests on these numbers.

## 3.1 What the rows prove

The signs come from the rows and hold wherever their hypotheses do. The dividend barrier rises in the asymmetry parameter and in the fixed issuance cost (M1). The recapitalisation trigger rises in the asymmetry parameter and falls in the fixed issuance cost, and the injection target and the gap from trigger to target rise in the fixed issuance cost (M2). The determinant of the two thresholds in the two parameters is negative by these signs alone (M3). The injection target does not depend on where an issue starts (M7), at $\lambda_S=1$ the model is the model with issuance costs alone (M8), and where the exposure binds the cap buffer variance is the limit at each date's asset volatility (M15). The hypotheses are listed in Appendix B.

Two further results come from the solve and are not rows. Above the reference the shortfall penalty is zero, so between the injection target and the dividend barrier the equation holds neither the asymmetry parameter nor the fixed issuance cost. There $V$ is fixed by the dividend barrier through $V'(y^*)=1$, $V''(y^*)=0$ and the equation at $y^*$, so wherever the injection target lies above the reference it is a function of the dividend barrier and the proportional

issuance cost and carries no information of its own about the two parameters. Required capital enters the shortfall penalty, through $R = 1 + q$, and not the regulatory cap, so the homogeneity behind Result 13 of Milne and Whalley [2001], by which a higher minimum raises desired capital one for one, fails here (L69, Section 3.4).

### 3.2 The baseline

The baseline is the parameter set of Bayraktar et al. [2026], so that every comparison with their Table 1 and their frontier is made at their numbers. They call it an illustrative parameterisation rather than a calibration (L31), and the paper takes it the same way. The riskless rate is $r = 0.02$, the expected asset return $\mu_s = 0.04$, the liability drift $\mu_L = 0.03$, the net income on liabilities $\gamma = 0.02$, the discount rate $\rho = 0.12$, the asset volatility $\sigma = 0.08$, the liability volatility $\sigma_L = 0.03$ and the return-liability correlation $c = 0.20$. The solvency floor $a_1 = 0.045$ is the CET1 minimum of 4.5 percent of risk-weighted assets, the run-off rate $a_2 = 0.05$ is the standard's factor for stable retail deposits, and the liquidity haircut $a_3 = 0.30$ is theirs and none of the standard's (Appendix C). The proportional issuance cost is $\kappa = 0.01$. The reference $R = 1.15$, required capital of 15 percent of liabilities, is the solver's choice rather than a regulatory number, and the sweep of required capital in Section 3.4 varies it from 1.05 to 1.25. The two parameters of K1 are set at $(\lambda_S, K) = (1.2, 0.05)$, a point at which the trigger is interior and every hypothesis of M1 to M4 holds.

The discount rate is the parameter the results depend on most, as it is for Milne and Whalley [2001], since it sets both the shareholders' impatience and, through $\rho_L = \rho - \mu_L$, the cost of holding a buffer. The value is finite only above $r + (\mu_s - r)/\max(a_1, a_3) = 0.087$. From $\rho = 0.10$ to 0.20 the dividend barrier falls from 2.18 to 1.51 and the trigger from 1.050 to 1.015, while the determinant of M3 stays between $-0.35$ and $-0.18$ and every hypothesis holds. At $\rho = 0.09$ the barrier reaches the top of the grid and the point is not used. The levels move with the discount rate and the signs do not.

### 3.3 The frictions

Figure 2 plots the three thresholds against the asymmetry parameter for three fixed issuance costs. At $K = 0.05$ a rise of $\lambda_S$ from 1 to 3 moves the dividend barrier from 1.810 to 1.900, the recapitalisation trigger from 1.025 to 1.083 and the injection target from 1.586 to 1.665, and the trigger is interior at all 33 points. The penalty charges each moment the state spends below the reference, so the bank shortens that time by issuing earlier and holds more before it pays out. The responses flatten as $\lambda_S$ rises. By M1 the response of the barrier is $\rho_L W(y^*)/(\rho_L - \mu'(y^*))$, and the expected discounted shortfall at the barrier falls from 0.062 at $\lambda_S = 1.05$ to 0.005 at $\lambda_S = 3$, since the bank that weighs the shortfall more spends less time in it. The determinant of M3 falls in size with it, from $-0.36$ to $-0.05$, with every hypothesis holding, so the identification of K1 is weaker at a large asymmetry parameter than at a small one. Finite differences on 601 nodes scatter around the formula, since the thresholds move by a node or two, and the formula is the reading used.

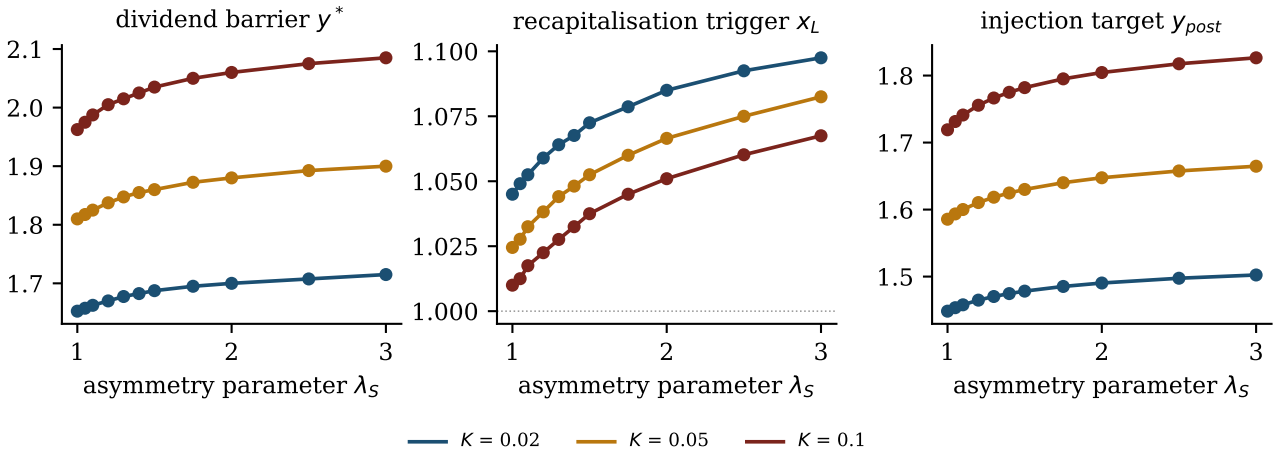



Figure 2: The dividend barrier, the recapitalisation trigger and the injection target against the asymmetry parameter, for $K = 0.02$, 0.05 and 0.1, at the baseline of Section 3.2. Filled markers have an interior trigger and open markers a trigger at the distress boundary. Source in Table 6.

Figure 3 plots the same thresholds against the fixed issuance cost. At $\lambda_S = 1.2$ a rise of $K$ from 0.005 to 0.15 moves the dividend barrier from 1.508 to 2.113 and the trigger from 1.086 to 1.015, so the gap from trigger to target widens from 0.24 to 0.84. A fixed cost per issue makes issues larger and rarer, as in Peura and Keppo [2006] (L22). At $\lambda_S = 1$ and $K = 0.15$ the trigger reaches the distress boundary, which is where K1 stops applying, since a trigger at the boundary cannot move.

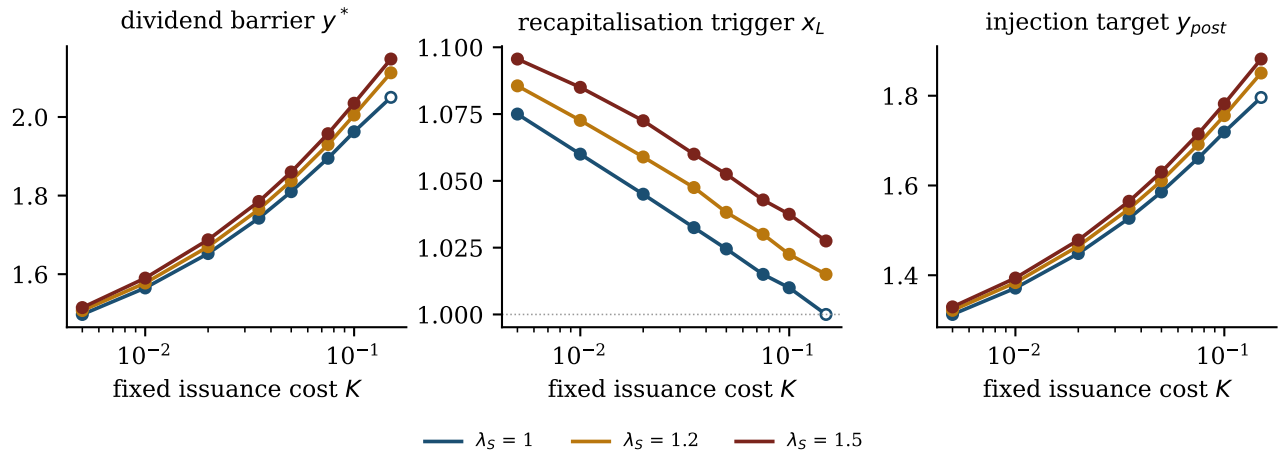



Figure 3: The three thresholds against the fixed issuance cost, on a log scale, for $\lambda_S = 1$, 1.2 and 1.5. Markers as in Figure 2. Source in Table 6.

The script `theory/solve/run_q1_frictions.py`, whose output is `theory/solve/results_q1.md`, moves one friction at a time around $(\lambda_S, K, \kappa) = (1.2, 0.05, 0.01)$. With the cap of A2 alone the trigger is interior at all 21 points, and at every one the dividend barrier rises with $\lambda_S$ and with $K$ while the recapitalisation trigger rises with $\lambda_S$ and falls with $K$, the signs of M1 and M2. The proportional issuance cost $\kappa$ lowers the trigger as $K$ does and lowers the injection target where $K$ raises it, so of the three frictions only $\lambda_S$ raises the trigger. Issues grow larger with $K$, as in Peura and Keppo [2006] (L22), and smaller with $\kappa$. With no borrowing the trigger reaches the distress boundary at $\lambda_S \leq 1.1$ and at $K \geq 0.1$, and at the 16 interior points the signs of M1 and M2 hold.

The script `theory/solve/run_q1_pk.py`, whose output is `theory/solve/results_q1_pk.md`, takes up three results of Peura and Keppo [2006] that the figures leave out. They pin their two frictions by the expected time between issues and the proportional cost of an issue (their Table 2). In the paper's model the time between issues rises with $K$ from 2.5 years at $K = 0.01$ to 9.8 years at $K = 0.1$ at $\lambda_S = 1.2$, and the cost per issue rises from 0.042 to 0.146, while $\lambda_S$ from 1 to 2 moves them from 6.8 to 6.0 years and from 0.099 to 0.096 at $K = 0.05$. The determinant of the two moments in $(\lambda_S, K)$ is non-zero at four centres, from $-0.48$ to $-0.86$, so the moments also separate the two frictions locally. Both moments fall in $\lambda_S$ and rise in $K$, so their determinant is the difference of two terms of the same sign and is set by the sizes of the responses. The determinant of the two thresholds is the difference of two terms of opposite sign, and is negative by the signs of M1 and M2 alone (M3), where the table of Peura and Keppo [2006] changes sign with the size of the costs (L41 to L43). The time between issues is read to about ten percent, which the header of the output explains.

The value of the option to issue is hump-shaped in the capital buffer in their Figure 4 and goes to zero as the buffer does, since a bank near the minimum rarely survives the delay. In the paper's model the value function with the option less the value function without it falls throughout the buffer at each of twelve combinations of $\lambda_S$ and $K$, and is largest at the distress boundary. At $(\lambda_S, K) = (1.2, 0.05)$ it is 0.40 there, 88 percent of the value function, against 15 percent at the injection target and 12 percent at the dividend barrier. It rises with $\lambda_S$, from 0.20 to 0.98 at $\lambda_S$ of 1 and 2, and falls with $K$. The hump needs the delay and liquidation, which the paper's model does not hold (Table 8), and here the option buys relief from the shortfall penalty and not survival.

Their Figure 6 and Table 3 find that the dividend barrier is driven by return volatility, with a correlation of 0.99 across banks. In the paper's model the dividend barrier rises with the asset volatility, from 1.44 to 2.16 as it goes from 0.04 to 0.12 at $\lambda_S = 1$, the recapitalisation trigger falls from 1.043 to 1.010, and the determinant of M3 grows in size from $-0.10$ to $-0.43$ with every hypothesis holding. At an asset volatility of 0.16 the barrier reaches the top of the grid, and that point is not used. The dividend barrier also rises with the expected asset return, from 1.19

to 2.04 as it goes from 0.025 to 0.045 at $\lambda_S = 1$, where theirs has a correlation of $-0.48$ with it across banks. A correlation across banks is not a comparative static, and the two are not set against each other.

The script `theory/solve/run_q1_delay.py`, whose output is `theory/solve/results_q1_delay.md`, asks whether the delay on issues of Peura and Keppo [2006] (L23) is a rival to K1. It adds the delay to the model in `theory/solve/delay_model.py`. The bank decides to issue at the recapitalisation trigger and the capital arrives a fixed time later, and while it waits it pays no dividend and bears the shortfall penalty. The state cannot fall below the distress boundary, so here the delay costs the penalty and the discounting over the wait, where in their model it also carries a risk of failure. On 1201 nodes, with thresholds accurate to about $3 \times 10^{-4}$ against 2401, a delay raises the dividend barrier, the recapitalisation trigger and the injection target at each of the four centres of `theory/solve/run_q1_pk.py`, as it raises both barriers in their Table 2 (`PeuraKeppo.table2_signs`).

The delay moves the dividend barrier and the recapitalisation trigger the way the asymmetry parameter does. The model without the delay, fitted to the two thresholds of the model with a delay of a tenth of a year, overstates $\lambda_S - 1$ by 9 to 78 percent across the four centres and the fixed issuance cost by 3 to 10 percent. A delay of a quarter of a year overstates $\lambda_S - 1$ by 27 to 260 percent. The fit is exact to within $4 \times 10^{-4}$ in both thresholds, so nothing in the two thresholds shows that a friction is missing.

The injection target does not show it either. In all twelve fits the fitted model's injection target is within $10^{-4}$ of the injection target of the model with the delay, below the accuracy of the solve. The injection target lies above the reference at every centre, where the shortfall penalty is zero, so from the injection target to the dividend barrier the equation holds none of the asymmetry parameter, the fixed issuance cost and the delay. There $V$ is fixed by the dividend barrier through $V'(y^*) = 1$, $V''(y^*) = 0$ and the equation at $y^*$, and the injection target, where $V' = 1 + \kappa$, is a function of the dividend barrier and the proportional issuance cost. It carries information about $\kappa$ and none about the three frictions. This argument is not a row, and it needs the injection target above the reference.

So the thresholds identify the asymmetry parameter and the fixed issuance cost within the model of A1 to A6 (M3), and with a delay on issues added they identify the asymmetry parameter only jointly with the delay. The two thresholds and the injection target are three numbers for four frictions, and the injection target goes to the proportional issuance cost. This bears on K1, whose identification assumes no delay, on C6, since the friction of Peura and Keppo [2006] is one the thresholds cannot tell from the shortfall penalty, and on L8, where they pin the delay and the fixed issuance cost from issue cost and frequency. Whether those issue moments separate the delay from the asymmetry parameter is not checked. No row rests on this pass. K1's scope note and C6 say that the identification assumes no delay.

### 3.4 Regulation

Figure 4 plots the dividend barrier, the trigger and the determinant of M3 against the solvency floor. At $\lambda_S = 1.2$ a rise of the floor from 0.045 to 0.12 moves the dividend barrier from 1.838 to 2.113 and the trigger from 1.038 to 1.123, and the determinant shrinks from $-0.29$ to $-0.04$ on 1201 nodes, with every hypothesis holding. A higher floor cuts the exposure the bank may take near distress, so the bank spends less time in deficit, the expected discounted shortfall at the barrier falls from 0.037 to 0.0007, and the asymmetry parameter has little left to act on. The sign of the determinant that M3 proves is kept, and the identification of K1 is weaker under a high floor than under a low one.

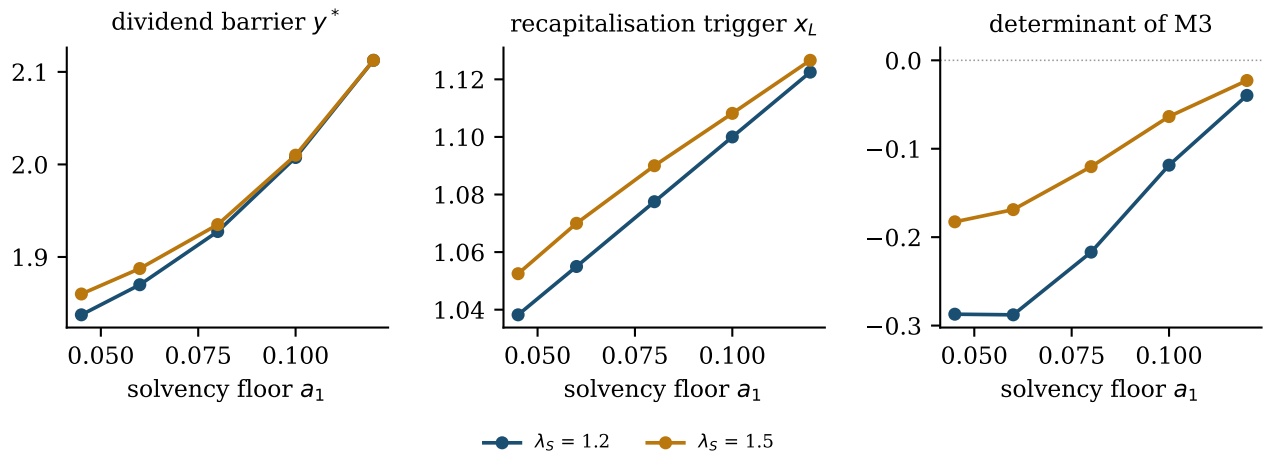



Figure 4: The dividend barrier, the recapitalisation trigger and the determinant of M3 by its formula against the solvency floor, for $\lambda_S = 1.2$ and 1.5 at $K = 0.05$. Source in Table 6.

Required capital moves the thresholds by less than itself. A rise of the reference from 1.05 to 1.25 raises the dividend barrier from 1.810 to 1.885, about three-eighths of the rise, and the trigger from 1.025 to 1.054, about a seventh, so the distance of the barrier above the reference falls from 0.76 to 0.64. In Milne and Whalley [2001] the same rise would move desired capital one for one and leave the buffer above the minimum as it was, since their equation is homogeneous in capital less the minimum (L69). Here required capital sets where the penalty starts and does not move the cap, whose branches start at the distress boundary, so a higher requirement narrows the buffer the bank holds above it.

The comparison with Table 1 of Bayraktar et al. [2026] rests on `theory/solve/run_q3_regulatory.py`, whose output is `theory/solve/results_q3.md`. A port of the authors' solver, `theory/solve/bcvw_port.py`, reproduces all 27 rows of their Table 1 to within $5 \times 10^{-5}$, and every row stops at the solver's limit of 1500 steps rather than at its tolerance. Solved by policy iteration in `theory/solve/bcvw_model.py`, the baseline dividend barrier is 1.389 against 1.428 printed and the value function at $y = 1.2$ is 0.870 against 0.855, while every direction in their table is unchanged. Their solver lets the bank issue at every state, and when converged the solver issues below 1.126 rather than at their distress threshold of 1.02. With issuance allowed only at 1.02, as their model is stated, the bank below 1.079 pays a dividend down to 1.02 and issues at once, and the slope of the value function rises from 1 to 1.038 below the injection target, so the value function is not concave at their baseline, against their Proposition 3.3. The early issuance and the rising slope hold on grids of 601, 1201 and 2401 nodes (`theory/solve/run_q3_concavity.py`), and the early issuance appears under their scheme and under policy iteration alike. Neither is proved, and the step of their proof that fails is not located. Their own remark that a fixed issuance cost could induce earlier issuance (p. 7) points the same way.

The paper's model reaches their numbers as the fixed issuance cost vanishes, `theory/solve/run_q3_pass2.py`. With $\lambda_S = 1$ and $K = 0.001$ the dividend barrier is 1.400 and the value function at 1.2 is 0.867, against 1.389 and 0.870 in their model converged, so the fixed issuance cost accounts for most of the gap between the two models' levels. The injection target still differs, 1.227 against 1.158, with the form of the proportional cost. At $(\lambda_S, K) = (1.2, 0.05)$ the dividend barrier, the injection target and the value function move with the cap geometry in the directions of their table, and every hypothesis of M1, M2 and M4 holds at each of the 17 points. The determinant of M3 does not hold its size, and `theory/solve/run_q3_pass3.py` confirms this on grids of 1201 and 2401 nodes. On the finer grid the formula of M3 evaluated on the solve gives a determinant that shrinks from $-0.289$ to $-0.041$ as the solvency floor rises from 0.045 to 0.12 (Table 5), and finite differences agree with it to within 15 percent at every point. The response of the dividend barrier to $\lambda_S$ is $\rho_L W(y^*)/(\rho_L - \mu'(y^*))$ by M1, and the expected discounted shortfall $W(y^*)$ falls from 0.039 to 0.0007 over the same range, while the responses to $K$ stay of order one. The sign that M3 proves is kept, and the identification of K1 is weaker under a high solvency floor than under a low one.

Table 5: Responses of the thresholds and the determinant of M3 along the cap geometry, by the formulas of M1 to M3 evaluated on the solve, on 2401 nodes, in the paper's model at the parameters of Bayraktar et al. [2026] with $(\lambda_S, K) = (1.2, 0.05)$.

| Cap geometry | $\partial y^*/\partial \lambda_S$ | $\partial x_L/\partial \lambda_S$ | $\partial y^*/\partial K$ | $\partial x_L/\partial K$ | Determinant of M3 | $W(y^*)$ |
|---|---|---|---|---|---|---|
| $a_1 = 0.045$ (baseline) | 0.104 | 0.057 | 4.27 | $-0.44$ | $-0.289$ | 0.0385 |
| $a_1 = 0.08$ | 0.041 | 0.054 | 3.47 | $-0.82$ | $-0.221$ | 0.0152 |
| $a_1 = 0.12$ | 0.002 | 0.018 | 2.15 | $-1.51$ | $-0.041$ | 0.0007 |
| $a_3 = 0.48$ | 0.144 | 0.063 | 1.98 | $-0.30$ | $-0.168$ | 0.0933 |

Which branch of the cap carries each result is a separate question from how the value loss divides, which is the question of Bayraktar et al. [2026]. `theory/solve/run_q3_regimes.py`, whose output is `theory/solve/results_q3_regimes.md`, solves the model under the ceiling $\pi \leq 4$ alone, with the solvency branch beside it, with the liquidity branch beside it, and with both, which is A2. The ceiling lies below the bound of 5 on the long-run exposure that keeps the value finite, so the four sets of admissible exposures are nested and every value is finite. The liquidity branch alone and the ceiling alone leave the exposure positive at the distress boundary, so they are devices for the decomposition and not regulations in force.

K1 needs the solvency branch. With it, whether or not the liquidity branch is present, the exposure is zero at the distress boundary, the recapitalisation trigger is interior at $\lambda_S$ of 1, 1.2 and 1.5, and it moves with $\lambda_S$ by 0.022 across that range under the solvency branch alone and by 0.029 under both. The determinant of M3 is $-0.33$ under the solvency branch alone and $-0.29$ under both. Without it the trigger is at the distress boundary at every $\lambda_S$ and the hypotheses of M1 to M4 fail, so the two thresholds do not identify $\lambda_S$ and $K$ there. The deficit limit of K2 needs the solvency branch as well. Buffer variance at the trigger is the deficit limit $v_1^2 = 3.14$ with it and is unbounded at the distress boundary without it. The surplus limit is not reached in these solves, since the dividend barrier lies between 1.7 and 2.3 in the four regimes, where buffer variance is between 0.31 and 0.36 against a surplus limit of 0.069, so its assignment to the liquidity branch rests on M11 and not on the solve. With no borrowing the solution is the same with and without the liquidity branch, to numerical precision, as C3 states.

### 3.5 Attitude to risk

Figure 5 plots the attitude to risk $-V''/V'$ across the inaction region, the measure Milne and Whalley [2001] plot in their Figure 4, near the trigger at full scale and over the whole region at a scale that shows the reference. The bank is averse to risk around the reference, where $-V''/V'$ is largest, between 0.77 and 0.88, and seeks risk just above the trigger, where $V$ is convex (M4) and $-V''/V'$ is between $-21$ and $-12$. A larger asymmetry parameter raises the aversion at the reference and damps the convexity at the trigger, from $-21$ at $\lambda_S = 1$ to $-12$ at $\lambda_S = 2$. At this baseline the exposure is at the cap over the whole inaction region at each of the four values, whatever the sign of $V''$ (`theory/solve/results_cs.md`), since the premium on the risky asset makes the interior maximiser exceed the cap. The attitude to risk therefore sets no exposure here, and the cap does. Their bang-bang choice of volatility by the sign of $V''$ (L62, L63) has no counterpart at this baseline, and nor has the shift towards riskier assets under a higher requirement that they report for concave utilities (L67), which Calem and Rob [1999] find in a dynamic bank that cannot issue (L72), since here a higher floor lowers the exposure through the cap. Their severely undercapitalized bank takes the most risk it can (L71), and a charge below required capital makes it take more, since it slows recapitalisation from earnings (L73). The paper's bank issues at the trigger instead, the shortfall penalty raises the trigger (M2), and the cap bounds the exposure where $V$ is convex, which is also where Fries et al. [1997] locate moral hazard (L77). Without an option to issue the value function of Milne and Robertson [1996] is concave all the way down to liquidation and aversion to risk rises as cash falls (L83), so the convexity just above the trigger comes from the option to issue.

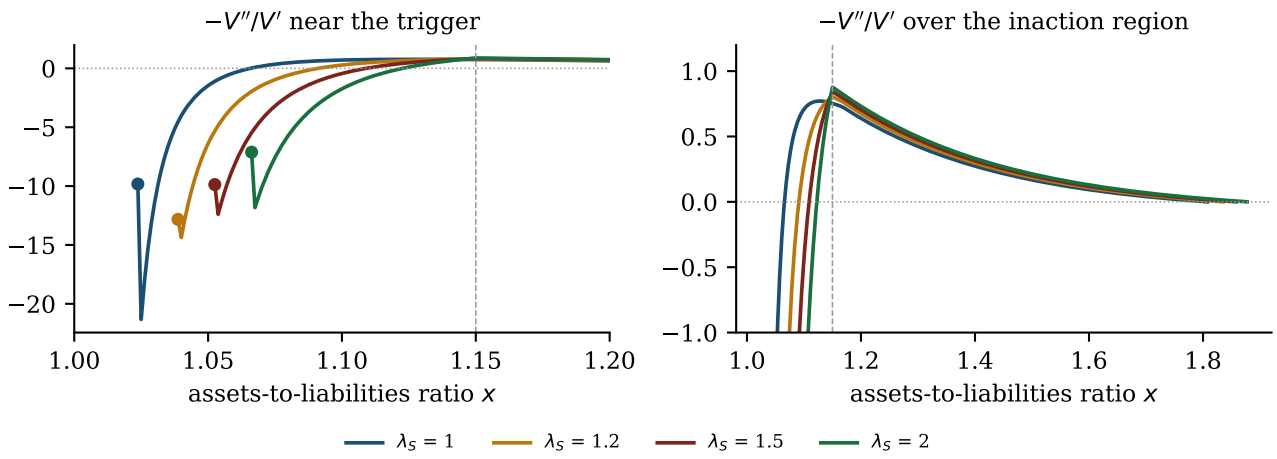



Figure 5: The attitude to risk $-V''/V'$ near the recapitalisation trigger and over the inaction region, for $\lambda_S = 1$, 1.2, 1.5 and 2 at $K = 0.05$. The dashed line is the reference and the markers are the triggers. Source in Table 6.

### 3.6 Safety and value

The frontier of Bayraktar et al. [2026] sets the value function against the probability of not reaching their distress threshold of 1.02 within five years from $y = 1.2$. In the paper's model the trigger lies above 1.02 at every point solved, so that probability is one throughout, and the safety measure used is the expected time in deficit over the same five years, with the expected shortfall, the probability of no deficit and the issues beside it. `theory/solve/q7_paths.py` simulates the optimal policy, under the physical measure for safety and under the measure of the value equation for values. Under the second it reproduces $V(1.2)$, $W(1.2)$ and $N(1.2)$ of the value equation to within about two percent, the size of the value equation's own grid error. Value is read two ways, as the value function and as the cash value $C = V + (\lambda_S - 1)W$, the dividends less issue costs with the penalty added back, since the value function holds the penalty and falls in $\lambda_S$ by construction.

Figure 6 sets the two levers side by side. A rise of the asymmetry parameter from 1 to 3 at $K = 0.05$ cuts the expected time in deficit from 0.79 to 0.14 years and the expected shortfall by 92 percent, at a cost of 4 percent of the cash value, and the bank gets there by issuing earlier and more often, 1.0 to 1.4 issues in five years. A rise of the solvency floor from 0.045 to 0.12 cuts the time in deficit to 0.12 years at a cost of 20 percent of the cash value. At the floor of 0.10, with 0.37 years in deficit, the same safety reached through the asymmetry parameter, at about $\lambda_S = 1.45$, leaves the cash value 0.082 higher and the value function 0.069 higher. Between floors of 0.045 and 0.06 the time in deficit rises, from 0.79 to 0.81 years, so a small rise of the floor buys no safety at all. Under the reading of Appendix D, in which the shortfall penalty stands for the supervisor's restrictions below required capital, a graded penalty buys safety more cheaply than a higher floor at this baseline. The comparison is of two levers at one baseline and is not a welfare statement.

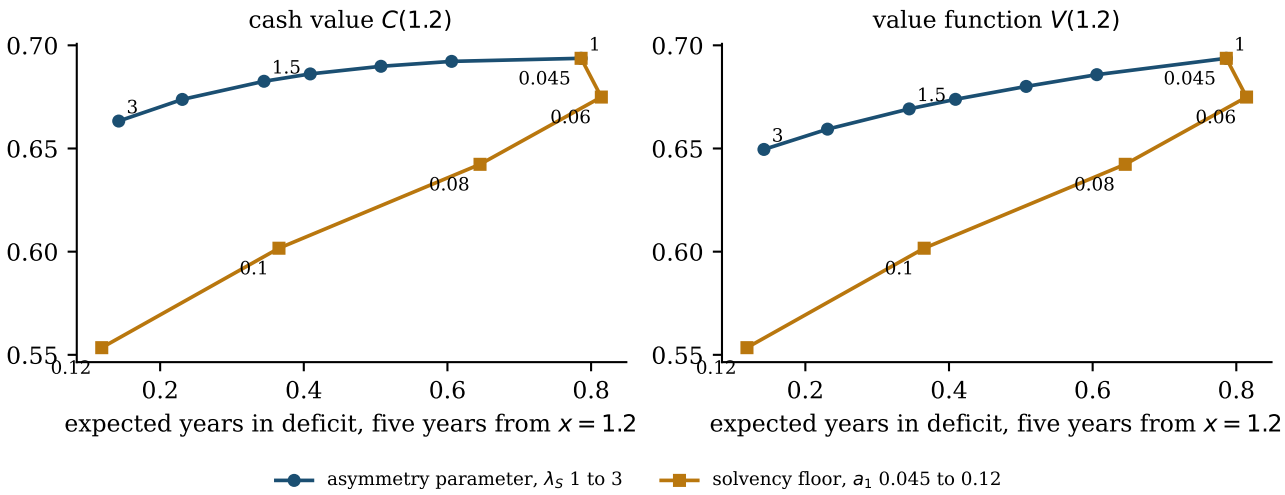



Figure 6: Cash value and value function at $x = 1.2$ against the expected years in deficit over five years from $x = 1.2$, along the asymmetry parameter from 1 to 3 at a solvency floor of 0.045 and along the solvency floor from 0.045 to 0.12 at $\lambda_S = 1$, both at $K = 0.05$. Source in Table 6.

### 3.7 Accuracy and what is not checked

The thresholds of the figures are on 601 nodes and accurate to about $10^{-3}$, the determinants on 601 and 1201 nodes, and the determinants of Table 5 on 2401. The simulated statistics have a standard error of at most 0.0035 for a probability. Three questions stay open. A model with time-varying asset volatility, whose thresholds would be curves in it, is not solved, and M15 covers only buffer variance where the cap binds. A delay on issues with liquidation while the issue is pending, the channel of Peura and Keppo [2006], is not solved. Whether the issue moments separate the delay from the asymmetry parameter is not checked.

### 3.8 Provenance

Table 6 gives, for each figure, table and group of numbers of the section, the script that writes it, the input it reads and the file it writes, with the first sixteen hexadecimal digits of that file's sha256. The figures are drawn from the saved outputs, so they are redrawn in seconds, and MS-11 of `theory/checks/verify_manuscript.py` redraws them into a temporary folder and checks that every file matches its digest. The outputs of the solves are not rerun by the check. Each states the sha256 of the scripts that wrote it in its header, and the commands that rerun them are in the scripts' docstrings.

Table 6: Provenance of the figures, tables and numbers of Section 3.

| Item | Written by | Reads | File | sha256 |
|---|---|---|---|---|
| Figure 2 | `theory/solve/make_cs_figures.py` | `theory/solve/results_cs.json` | `theory/figures/cs_lambda.pdf` | `c2b32a8b2919e74b` |
| Figure 3 | `theory/solve/make_cs_figures.py` | `theory/solve/results_cs.json` | `theory/figures/cs_K.pdf` | `07d0b19ad757f742` |
| Figure 4 | `theory/solve/make_cs_figures.py` | `theory/solve/results_cs.json` | `theory/figures/cs_floor.pdf` | `ea4a6b4e8dcfd1b5` |
| Figure 5 | `theory/solve/make_cs_figures.py` | `theory/solve/results_cs.json` | `theory/figures/cs_risk.pdf` | `e6039824c2f11ac1` |
| Figure 6 | `theory/solve/make_cs_figures.py` | `theory/solve/results_q7.json` | `theory/figures/cs_frontier.pdf` | `54ca5cf436f01ba5` |
| Levers at equal safety, Section 3.6 | `theory/solve/make_cs_figures.py` | `theory/solve/results_q7.json` | `theory/solve/results_cs_frontier.md` | `dbc2defc54f9c4b1` |
| Data of Figures 2 to 5 | `theory/solve/run_cs.py` | `theory/solve/solve_qvi.py` | `theory/solve/results_cs.json` | `f40b4d5b23f82bee` |

| Item | Written by | Reads | File | sha256 |
|---|---|---|---|---|
| Numbers of Sections 3.2 to 3.5 | theory/solve/run_cs.py | theory/solve/solve_qvi.py | theory/solve/results_cs.md | 9ca61dab60e89348 |
| Data of Figure 6 | theory/solve/run_q7_safety.py | theory/solve/q7_paths.py | theory/solve/results_q7.json | eb8ef952e5cdbc7d |
| Numbers of Section 3.6 | theory/solve/run_q7_safety.py | theory/solve/q7_paths.py | theory/solve/results_q7.md | 8403f3559d48f0b2 |
| Table 5 | theory/solve/run_q3_pass3.py | theory/solve/solve_qvi.py | theory/solve/results_q3_pass3.md | fbd271b583198235 |
| Table 1 of Bayraktar et al. [2026] | theory/solve/run_q3_regulatory.py | theory/solve/bcvw_port.py, theory/solve/bcvw_model.py | theory/solve/results_q3.md | 65cb3a8912c79726 |
| Their value function's concavity | theory/solve/run_q3_concavity.py | theory/solve/bcvw_model.py | theory/solve/results_q3_concavity.md | 33e0cf745c380e13 |
| The bridge as the fixed issuance cost vanishes | theory/solve/run_q3_pass2.py | theory/solve/solve_qvi.py | theory/solve/results_q3_pass2.md | 42e68d39b7b905fb |
| The branches of the cap | theory/solve/run_q3_regimes.py | theory/solve/solve_qvi.py | theory/solve/results_q3_regimes.md | c6fc9c71d863fb21 |
| One friction at a time | theory/solve/run_q1_frictions.py | theory/solve/solve_qvi.py | theory/solve/results_q1.md | 4251f98f1161e71e |
| Issue moments, option value, return moments | theory/solve/run_q1_pk.py | theory/solve/solve_qvi.py | theory/solve/results_q1_pk.md | 8f566e271a8ceef0 |
| A delay on issues | theory/solve/run_q1_delay.py | theory/solve/delay_model.py | theory/solve/results_q1_delay.md | 3dce10e25314d214 |

# 4 Literature

## 4.1 How the papers read leave the question open

Among the dynamic models of bank capital read, each makes part of the bank's policy endogenous and fixes the rest. Milne and Robertson [1996], the base of the bank papers, make the dividend barrier endogenous for a firm that cannot issue and is liquidated at a threshold (L81). In Milne and Whalley [2001] the dividend barrier, which is also the target of each issue, and the volatility the bank bears are endogenous, and the bank issues only when a random audit finds it below the minimum (L38, L62, L64). Peura and Keppo [2006] make the dividend barrier and the issue barrier endogenous, with a single target, under a delay on issues and liquidation at the minimum, and the bank's risk is given (L7, L23, L24). Dai et al. [2019] keep that structure and add partial information about the bank's assets (L9, L25). Bayraktar et al. [2026] make the dividend barrier, the injection target and the risky exposure under a regulatory cap endogenous, and fix the state at which the bank must recapitalise (L1, L28). Federico et al. [2026] put the issue trigger at required capital, where it is not a choice (L60), and Glover et al. [2022] force an issue at a breach (L55). Calem and Rob [1999] make the risky share endogenous in a bank that cannot issue and rebuilds capital from its earnings (L71), and Fries et al. [1997] make closure endogenous, chosen by shareholders who inject capital continuously until it no longer pays (L75, L76). For firms with no regulator, Bolton et al. [2011] make the payout and financing barriers endogenous with a fixed issuance cost (L10), and Décamps et al. [2011] find that the same cost makes issues lumpy (L57). No model read makes the recapitalisation trigger and the risky exposure under a regulatory cap endogenous together. The paper's bank chooses its dividend barrier, its recapitalisation trigger, its injection target and its risky exposure under the cap (P1). K1 is a statement about the first three, and K2 about the fourth.

The asymmetry parameter and the fixed issuance cost stand for the two frictions that the papers read use to explain a buffer above the minimum, a cost of falling below the minimum and a cost of raising capital back. The bank papers read write the cost of falling below in six forms, and none of the six is a running cost charged only below the reference that grows with the shortfall, as the shortfall penalty of A4 is. Milne and Robertson [1996], whose firm the bank papers build on, name such a charge, a penalty rate on debt below their liquidation threshold, as an alternative to liquidation and do not solve it (L82). Below the minimum the bank is liquidated in Peura and

Keppo [2006] (L24), liquidated or made to issue at a fixed cost in Federico et al. [2026] (L59), audited at random in Milne and Whalley [2001] (L16), and made to issue at once at the proportional issuance cost in Glover et al. [2022], so that one parameter prices both the shortfall and the issue (L55). Below a buffer requirement Behn et al. [2019] and Lang and Menno [2026] charge an amount that is the same at every shortfall (L53, L51), and with that charge the bank of Lang and Menno [2026] holds capital exactly at the buffer requirement and no buffer above it (L51). Lambertini and Uysal [2014] write the capital requirement as a smooth penalty on net worth that charges a shortfall and rewards a surplus, with no kink at required capital (L56). The cost of raising capital is fixed per issue in Milne and Whalley [2001], Peura and Keppo [2006] and Décamps et al. [2011], where the fixed part is what makes issues lumpy (L57), and has no fixed part in Behn et al. [2019] and Glover et al. [2022] (L54, L55).

Where the papers read put a cost of falling below and a cost of raising capital in one objective, none asks whether the policy separates the two. Outside banking the pairing is the cash-management model of Constantinides and Richard [1978], a running penalty that grows linearly below a level together with a fixed and a proportional cost of each transfer (L48). In their approximation with no drift and vanishing discounting the upper trigger rises in both the penalty and the fixed cost of raising cash, the signs M1 gives for the dividend barrier, and they state no direction for the lower trigger (L49, L50). In a bank Milne and Whalley [2001] pair an audit at random below the minimum with a fixed cost of each issue, so that the capital decision is an inventory problem with a target (L37, L38), in which the bank issues only when an audit finds it below the minimum (L64). More frequent audit and a higher fixed cost of issue move the policy in the same direction (L39, L40). With like responses the signs of the two thresholds do not separate the two frictions, since the Jacobian can then be singular (M3). Behn et al. [2019] estimate a charge for breaching the capital requirement and an issuance cost with no fixed part in one bank, and separate the two by assigning a different moment to each rather than by the policy (L54). Federico et al. [2026] pair the fixed issuance cost with a payout tax that depends on the macro state rather than on the shortfall, and issue at required capital, which is not a choice, so that every response of their thresholds is numerical at one calibration (L60). Peura and Keppo [2006] pair the fixed issuance cost with a delay on issues rather than with a cost of falling below the minimum (L7, L23). The paper asks whether the policy separates the two frictions, with the cost of falling below the minimum written as a shortfall penalty that grows below required capital (A4).

The papers read settle the premises of the question and leave the question itself. Banks hold capital well above the regulatory minimum (L15), and what a buffer above the minimum guards against is a costly intervention by the supervisor or a response of the market once capital falls below the minimum (L16). An optimising bank with a fixed issuance cost and liquidation at the minimum underpredicts the level (L17). Peura and Keppo [2006] call that penalty extreme and ask for one that grows as the capital ratio deteriorates (L18); the shortfall penalty of A4 is that specification, with the asymmetry parameter $\lambda_S$ as the weight it puts on a shortfall. The asymmetry parameter must then be set, and the two papers that calibrate this class set their frictions by hand or from values. Peura and Keppo [2006] impose common values by intuition and pin the delay and the fixed cost from issue cost and frequency (L20, L8), and Dai et al. [2019] fit their parameters to equity values (L21). The asymmetry parameter matters because at one the shortfall penalty is off and the model is theirs (M8), so an identified asymmetry parameter is what would let the specification be tested rather than imposed. K1 answers that part (M1 to M3, C1, C6). Whose cost the penalty is, which Peura and Keppo [2006] name as the issue their omissions reduce to (L19), is not settled by the thresholds and stays outside the paper (Appendix D).

On buffer variance the papers read point both ways. In an unconstrained loss-averse control the large-wealth limits of the controls depend on the preference parameter (L14); in an endowment economy with constant loss aversion the volatility of returns equals that of dividends (L12); in capped portfolio insurance the two volatility levels are set by the multiple and the leverage limit, which preferences choose (L5, L6). In equity returns of financial firms Engle and Siriwardane [2018] write variance as the squared leverage multiplier times asset variance, using the relation for returns and not for levels (L34, L70), and find that mechanical leverage explains little of the asymmetry of equity volatility, most of which is exposure to the market (L13). In K2 the multiplier is the risky exposure over equity, which the cap fixes at the reciprocal of the solvency floor near the distress boundary and of the liquidity haircut deep in surplus, so the gap between the deficit limit and the surplus limit is leverage set by regulation, with the liability shock added, and the asymmetry parameter does not enter it. For a bank under a regulatory cap no paper read evaluates the limits. K2 does (M10 to M12, C2), and C3 says where the surplus limit applies (L2).

Table 7: The elements of the paper's ledger against the three approaches.

| Element of Figure 1 | Issuance-cost models | Regulatory-cap control | Preference in volatility | This paper |
|---|---|---|---|---|
| A fixed issuance cost that makes issues lumpy (A5, M6) | present (L7, L22) | absent (L3) | absent | present |
| A shortfall penalty growing below required capital (A4, M8) | liquidation instead, a graded penalty called for (L24, L18) | an exogenous threshold (L28) | a preference over losses, no bank (L12) | present, with $R = 1 + q$ |
| A regulatory cap on exposure (A2, M10) | absent | present (L26) | absent | present |
| A recapitalisation trigger moved against the fixed issuance cost (M2, M3) | the regulator's asymmetry, through the liquidation barrier (L25) | the trigger is the distress boundary (L28) | no trigger | the asymmetry parameter (M2, M3) |
| Parameters read off the thresholds (M3) | set by calibration or fit (L8, L20, L21) | illustrative (L31) | read off variance (L12, L14) | local identification (M3) |
| Limits of buffer variance set by the regulatory cap, free of the asymmetry parameter (M11) | not evaluated | not evaluated, and preference-set slopes in the rule of Black and Perold (L6) | the reference parameter in the limits, with no cap (L14) | present (M11) |

## 4.2 The three approaches as ledgers

The papers read address the question in three ways, and each is drawn as a ledger of the same form as the paper's own (Figure 1), with the entities that act or are acted on and the actions between them. An arrow is solid when the paper derives the action and its Lean record proves it, dashed when the paper assumes it, and dotted when the paper states it in words and does not model it. Table 7 sets the elements of the paper's ledger against the three.

The first approach is the issuance-cost barrier models (Figure 7). The bank pays dividends at a barrier and issues in lumps because issues carry a fixed cost (L7, L22), and these thresholds and a fixed issuance cost per unit of size are the paper's (M1, M2, M7, A5). The shortfall is liquidation at the minimum (L24), which Peura and Keppo [2006] would replace with a penalty that grows as the ratio falls (L18), and in Dai et al. [2019] the asymmetry is the regulator's and acts through the liquidation barrier (L25). No parameter is read off the thresholds. The analyst pins them from issue cost and frequency, imposes them or fits them to equity values (L8, L20, L21), and that dotted arrow is where the paper's M3 enters.

The second is regulatory-cap control (Figure 8). The cap is the paper's (L1, L26), and the thresholds survive it (L28), since the cap moves them and does not create them (L29). The trigger is the exogenous distress boundary and issues carry proportional costs only, and Bayraktar et al. [2026] name a fixed cost as the change that would move recapitalisation (L3). There is no asymmetry parameter, and the parameters are illustrative (L31). The rule of Black and Perold [1992] has the same geometry with preferences in place of regulation (L5, L6) and a safe asset that can track the liability (L32), so its two volatility levels carry preference information. In the paper the slopes are regulatory, the liability shock is unhedged, and the limits carry no information about the asymmetry parameter (M11).

The third reads a preference asymmetry in volatility (Figure 9). The asymmetry is an investor's preference over gains and losses, and its mark is sought in the variance of returns. With constant loss aversion it leaves none (L12), the volatility it adds comes from loss aversion that varies with prior outcomes (L33), and without a cap the large-wealth limits of the controls carry the reference parameter (L14). Engle and Siriwardane [2018] assign volatility asymmetry to market exposure rather than to leverage (L13, L34). Under a binding regulatory cap the paper's limits carry no mark of the asymmetry parameter (M11).

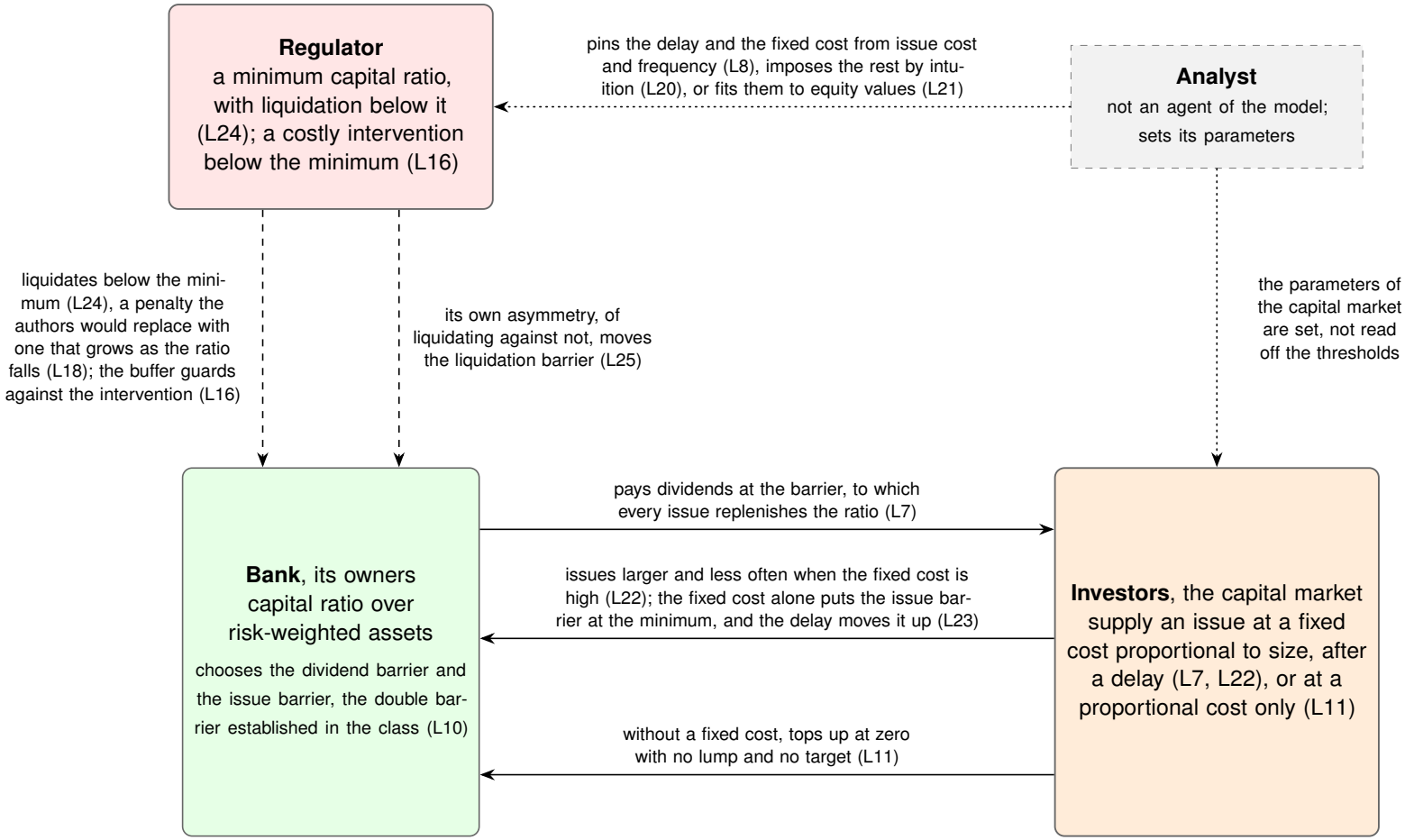



Figure 7: Issuance-cost barrier models, Peura and Keppo [2006], Dai et al. [2019], Bolton et al. [2011], Løkka and Zervos [2005] and Milne and Whalley [2001]. Solid arrows are derived in the papers and proved in their Lean records, dashed arrows are assumed, and dotted arrows are stated in words and not modelled. The analyst is not an agent of the models. Against Figure 1, there is no cap and no graduated penalty, and the parameters are set rather than read off the thresholds.

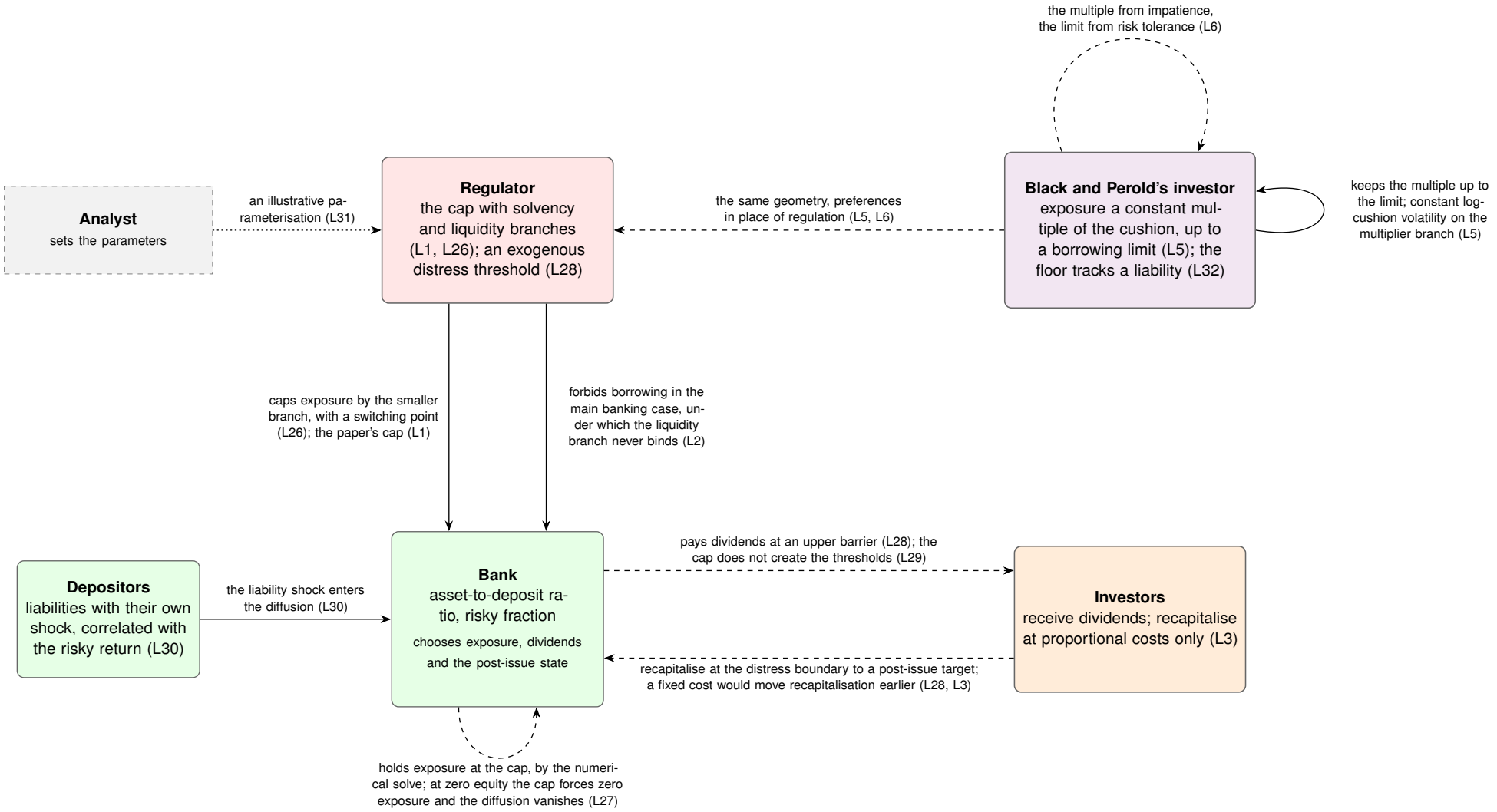



Figure 8: Regulatory-cap control, Bayraktar et al. [2026], with the rule of Black and Perold [1992] as the same geometry under preferences. Arrow styles as in Figure 7. Against Figure 1, there is no fixed issuance cost, the trigger is exogenous, and there is no asymmetry parameter.

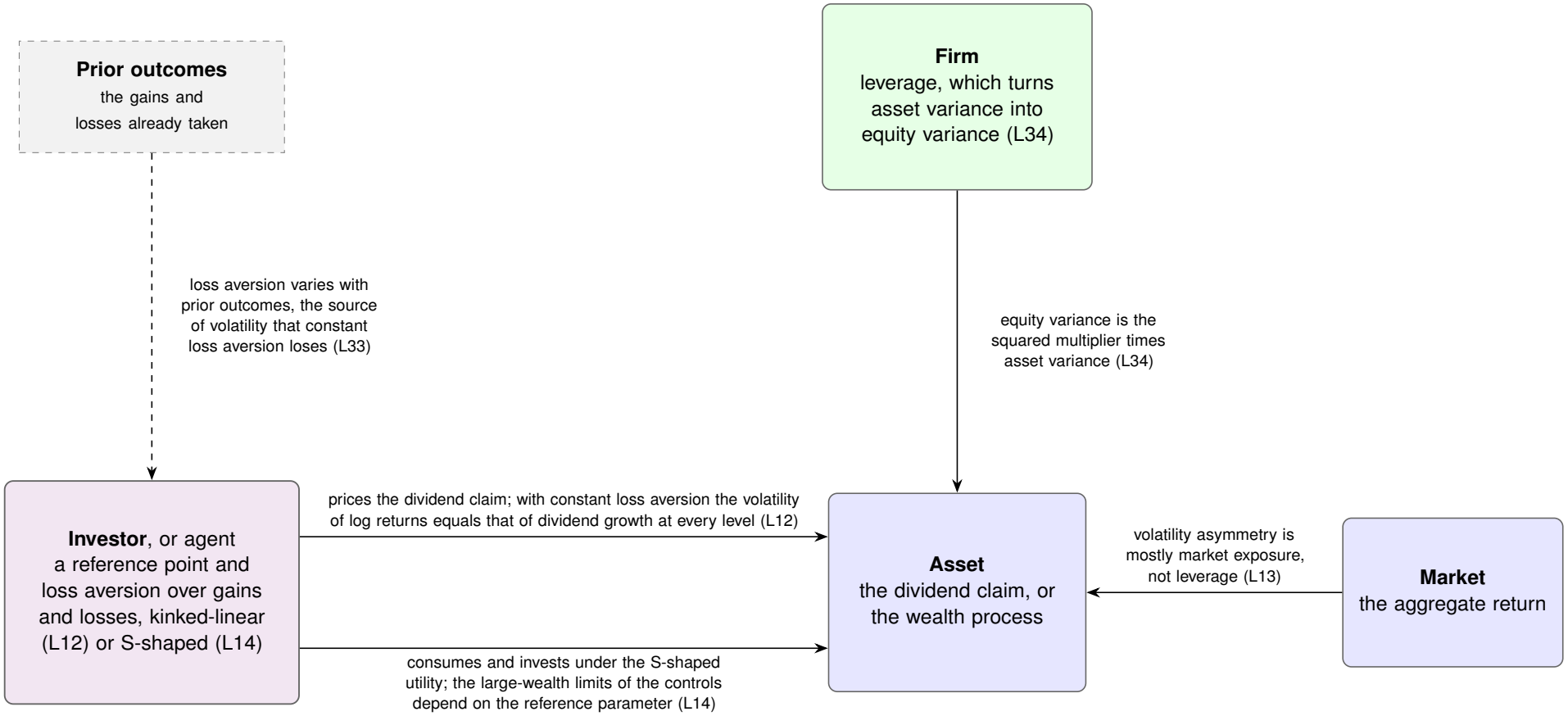



Figure 9: Preference asymmetry read in volatility, Barberis et al. [2001], Li et al. [2023] and Engle and Siriwardane [2018]. Arrow styles as in Figure 7. There is no bank and no regulator. Against Figure 1, the asymmetry is sought in the variance of returns, which a binding cap closes (M11), and not in thresholds.

### 4.3 The three approaches in one formulation

The paper's model is the problem of Section 2.3, whose equation (1) is one quasi-variational inequality, and the three approaches are the three places in it where a friction can sit. The regulatory cap of A2 is the set over which the supremum is taken, the cap term. The fixed and proportional issuance costs of A5 sit in the impulse operator, the impulse term. The asymmetry parameter sits in the running payoff, the payoff term. This is the equation that `theory/solve/solve_qvi.py` solves.

Each approach fills one of the three terms and leaves the other two at a restriction. Regulatory-cap control is the cap term, with $\lambda_S = 1$, $K = 0$ and the impulse operator evaluated at the distress boundary alone (L3, L28). The cap term also holds the control of variance of Milne and Whalley [2001], whose bank takes the smaller of two assumed volatilities where its value is concave and the larger where it is convex (L62, L63). The rule of the cap term reduces to theirs when the risk premium is zero, with no exposure and the regulatory cap in place of their two volatilities, so their bound on risk is assumed where that of Bayraktar et al. [2026] is set by regulation. The issuance-cost models are the impulse term, with no cap and $\lambda_S = 1$ (L7, L22). The preference approach is the kink of the payoff term, the loss-averse form of Barberis et al. [2001] written over a shortfall of capital rather than over gains and losses (L36). Table 8 gives the restriction for each paper read. Its four columns are the payoff term, the impulse term, the cap term and the rule at the distress boundary, and its last column names what the equation does not hold.

The payoff term holds more of the papers read than the kink alone. The charge that is the same at every shortfall in Behn et al. [2019] and Lang and Menno [2026] (L53, L51), the smooth penalty of Lambertini and Uysal [2014] (L56) and the linear penalty of Constantinides and Richard [1978] (L48) are each a choice of $\Lambda$, and the kink of A4 is one member of that family (M8). Liquidation at the minimum in Peura and Keppo [2006] and Federico et al. [2026] (L24, L59) and the forced issue of Glover et al. [2022] (L55) are not choices of $\Lambda$ but rules at the distress boundary, which the equation holds as a boundary condition there in place of the impulse operator. The delay on issues of Peura and Keppo [2006] (L23) and the partial information of Dai et al. [2019] are outside the equation. The random audit of Milne and Whalley [2001] is a payoff term of another kind. Below the minimum it charges, at the audit's rate, the value lost by being forced to issue or close, which depends on $V$ and so is not a choice of $\Lambda$. With a fine in proportion to the shortfall in place of forced action it is the shortfall penalty, with $\lambda_S - 1$ equal to the rate times the fine (L65). So is the preference of Black and Perold [1992], whose rule the cap term reproduces as a policy without its reason (L5, L6), and so are the equilibrium of Barberis et al. [2001] and the running maximum of Li et al. [2023] (L12, L14).

The structural GARCH model of Engle and Siriwardane [2018] is not a fourth approach, since no one in it chooses a policy. Their leverage multiplier turns asset volatility into equity volatility, is a function of observed leverage and rises in it (L79, L80), and their asset volatility is a GARCH process whose response to news they estimate (L13). In the paper's model the same multiplier is the risky exposure over equity, which the bank chooses and the cap fixes where it binds, at $1/a_1$ near the distress boundary and towards $1/a_3$ deep in surplus. With no liability shock buffer volatility is then asset volatility times the regulatory multiple (M14), at each date when the asset volatility moves (M15), so within each branch of the cap the leverage effect they measure is absent. The difference between the deficit limit and the surplus limit is a switch of the multiplier between the branches, which the asymmetry parameter does not enter (K2). The model implies their identity with the multiplier set by regulation, and says nothing of their estimation.

Table 9 evaluates two quantities of the equation on each paper read. One is the determinant of the two thresholds in the two frictions $\theta_1, \theta_2$ that an approach has,

$$J = \det \frac{\partial (y^*, x_L)}{\partial (\theta_1, \theta_2)},$$

read in the issuance-cost models with their dividend barrier and issue barrier in place of the dividend barrier and the recapitalisation trigger. The other is the pair of limits of buffer variance, $v_1^2$ and $v_3^2$, where the risky exposure binds the cap. Each value is at the paper's own parameters, so the comparison is of signs and of which quantity exists, rather than of sizes across papers.

The determinant separates the paper from the issuance-cost models. In the table of Peura and Keppo [2006] the delay and the fixed cost give $J = -29/100$ at small costs and $J = +1/25$ at the highest cost (L8, L41, L42, L43), so the two frictions are separated at each point of the table by the size of their responses and not by a sign that holds throughout it. In Dai et al. [2019] the delay and the fixed cost move the issue barrier the same way and $J \approx -0.0041$ (L9, L44), non-zero by magnitude. Milne and Whalley [2001] report like signs for the audit and the

fixed cost and print no values (L40), so $J$ cannot be formed. In the paper $J < 0$ under the hypotheses of M1 to M3 by the signs of the responses alone (M3), so the separation does not rest on a calibration.

The limits separate the paper from regulatory-cap control and from the preference approach. At the baseline of Bayraktar et al. [2026], which they do not evaluate, the deficit limit is about 3.140 and the surplus limit about 0.0688, a ratio of about 45.6 (M13, L45, L46), and without borrowing, their main banking case, the surplus limit is not reached (C3). With no liability shock the same limits are $256/81 \approx 3.160$ and $16/225 \approx 0.0711$, the log-cushion variances of capped portfolio insurance with multiples $1/a_1$ and $1/a_3$ (M14, L5), so at that baseline the unhedged liability shock lowers the deficit limit by about 0.6 percent and the surplus limit by about 3.2 percent. The bank of Dai et al. [2019] has no liability shock (L47), and in the rule of Black and Perold [1992] the safe asset can hedge it (L32) and the multiples are set by preferences (L6). In the unconstrained control of Li et al. [2023] the large-wealth limits depend on the reference parameter (L14), while in the paper neither limit depends on the asymmetry parameter (M11). In Barberis et al. [2001] constant loss aversion leaves the volatility of returns at that of dividend growth, 12 percent at every level (L12), free of the asymmetry because the price-dividend ratio is constant rather than because a cap binds.

The equation says where each result of the paper lives. The asymmetry parameter enters the payoff term alone and the cap enters the cap term alone. Where the cap binds, the exposure is $u(x)/x$ whatever $V$ is, so buffer variance is a function of $(\sigma, \sigma_L, c, a_1, a_3, x)$ and carries nothing of $\lambda_S$, which is M10 and M11. Where the cap does not bind, the exposure is the interior maximiser, which solves $\frac{1}{2}\,\partial_\pi \sigma^2(x,\pi)\,V'' + \partial_\pi \mu(x,\pi)\,V' = 0$ and so depends on $V'/V''$ and through it on $\lambda_S$. The solve with no borrowing has such an interval, where the exposure is below the cap and below the ceiling of one, and there buffer variance at $\lambda_S = 1.6$ differs from its value at $\lambda_S = 1.2$, while where the exposure is at the ceiling for both it is the same (`theory/solve/results_noborrow.md`, section 4). The two thresholds are set by the other two branches of the maximum, $V' = 1$ at the dividend barrier and $V = \mathscr{M}V$ at the recapitalisation trigger, and each branch involves the whole equation, so the thresholds carry $\lambda_S$ and $K$ together (M1 to M3). K1 is a statement about those two branches and K2 about the cap term, and the paper reads the asymmetry parameter off the thresholds rather than off buffer variance because the equation keeps it out of the term the cap pins.

Table 8: The three approaches as restrictions of the paper's equation.

| Paper | Payoff term | Impulse term | Cap term | Rule at the distress boundary | Outside the equation |
|---|---|---|---|---|---|
| Peura and Keppo [2006] | $\lambda_S = 1$ | $K > 0$ (L7) | none | liquidation at the minimum (L24) | the delay on issues (L23) |
| Dai et al. [2019] | $\lambda_S = 1$ | $K > 0$ (L9) | none | liquidation, with the regulator's asymmetry (L25) | the delay, partial information |
| Milne and Whalley [2001] | $\lambda_S = 1$ | $K > 0$ (L39) | $\sigma_1$ or $\sigma_2$, assumed (L62) | an audit at random below the minimum, the only time the bank issues (L64) | none, the audit is a payoff term that depends on $V$ (L65) |
| Bayraktar et al. [2026] | $\lambda_S = 1$ | $K = 0$, $\kappa > 0$ (L3, L46) | $u$ (L26) | issue or liquidate at an exogenous trigger (L28) | none |
| Black and Perold [1992] | no asymmetry | no issuance | $u$ with $a_2 = 0$, slopes from preferences (L5, L6) | none | preferences over consumption |
| Barberis et al. [2001], Li et al. [2023] | loss aversion over gains and losses (L36, L14) | no issuance | none | none | not a bank |
| This paper | $\lambda_S \geq 1$, $R = 1 + q$ (A4) | $K > 0$, $\kappa \geq 0$ (A5) | $u$ (A2) | the impulse operator, trigger endogenous (A6) | none |

Table 9: The two quantities of the equation, each at the paper's own parameters.

| Paper | Two frictions | Determinant $J$ of the two thresholds | Limits of buffer variance | The asymmetry enters |
|---|---|---|---|---|
| Peura and Keppo [2006] | delay, fixed cost | $-29/100$ at small costs, $+1/25$ at the highest (L8, L41) | not evaluated | nowhere |
| Dai et al. [2019] | delay, fixed cost | about $-0.0041$, like signs (L44) | not evaluated | the liquidation barrier (L25) |
| Milne and Whalley [2001] | audit, fixed cost | like signs, no values (L40) | not evaluated | nowhere |
| Bayraktar et al. [2026] | one cost | none, the trigger is exogenous (L28) | about 3.140 and 0.0688 at their baseline (M13), the second not reached without borrowing (C3) | nowhere |
| Black and Perold [1992] | none | no trigger | $256/81$ and $16/225$ at the same slopes, with no liability shock (M14) | the slopes, by preferences (L6) |
| Barberis et al. [2001], Li et al. [2023] | none | no trigger | return volatility 12 percent at every level (L12); limits that depend on the reference (L14) | the limits, without a cap (L14) |
| This paper | $\lambda_S$, $K$ | $J < 0$ by sign (M3) | free of $\lambda_S$ (M11) | the thresholds (M1 to M3) |

## 4.4 Papers read

Table 10: Papers read, with the passage each row relies on.

| ID | Paper | What we rely on | Quote | Page | Rows |
|---|---|---|---|---|---|
| L1 | Bayraktar et al. [2026] | The cap $u$ and the crossing point $\bar{x}$ of M10 are this paper's eq. (4) and its switching point. M10 and M11 are statements about its cap. The parallel ends there, since the paper evaluates no limit of the volatility and has no asymmetry parameter. | "the parameter a3 scales the entire liquidity constraint and determines its asymptotic upper bound" | 17 | M10, M11 |
| L2 | Bayraktar et al. [2026] | The authors' main banking case has no borrowing. With no borrowing the liquidity branch lies above the ceiling once $a_2 \leq (1-a_3)x$, so it never binds (`BCVW.ceiling_tighter_than_lcr`), and M11's surplus limit belongs to their unrestricted benchmark. | "we treat the $\pi \leq 1$ case as the main banking interpretation and the unrestricted case as a mathematical benchmark" | 19 | M11 |
| L3 | Bayraktar et al. [2026] | The paper has no fixed issuance cost, fixes its recapitalisation threshold exogenously, and names a fixed cost as a change that could move recapitalisation. M2 and M3 take up that change. | "If one introduced fixed issuance costs, endogenous market access, or adverse-selection costs that rise near distress, earlier recapitalization or inaction at the boundary could become optimal." | 13 | M2, M3 |
| L4 | Bayraktar et al. [2026] | A fixed issuance cost charged in dollars breaks the reduction to one dimension. M1 to M3 charge $K$ per unit of liabilities, which keeps it (Appendix D). | "Fixed issuance costs, nonlinear issuance costs, taxes, deposit insurance premia, endogenous deposit rates, balance-sheet size effects, multiple asset classes, or regulatory constraints depending on dollar levels rather than ratios would generally break the one-dimensional reduction." | 9 | M1, M2, M3 |
| L5 | Black and Perold [1992] | Their rule is the regulatory cap with $a_2 = 0$, multiple $1/a_1$ and leverage limit $1/a_3$ (`BlackPerold.cap_is_cppi`), and its log-cushion volatility on the multiplier branch is constant (their Prop. 2), the constancy of $v_1^2$ in M10. The parallel ends at the liability, which their reserve asset tracks and ours does not. | "maintains the portfolio's risk exposure a constant multiple of the excess of wealth over a floor, up to a borrowing limit" | 403 | M10, M11 |

| ID | Paper | What we rely on | Quote | Page | Rows |
|---|---|---|---|---|---|
| L6 | Black and Perold [1992] | In their Prop. 11 the multiple is set by the impatience to consume and the leverage limit by risk aversion, so the two volatility levels carry preference information. In M11 the slopes are regulatory and the asymmetry parameter is absent from the limits. | "The multiple depends on the utility function only through" | 421 | M11 |
| L7 | Peura and Keppo [2006] | The bank form of the class, with a fixed issuance cost as a fixed proportion of size, the device of M1 to M3, and each issue ending at the dividend barrier, a single target as in M7. | "As in the general case, the dividend barrier is also the level up to which the capital ratio is replenished each time a new capital issue is implemented." | 2176 | M1, M2, M7 |
| L8 | Peura and Keppo [2006] | They pin the issue delay and the fixed cost by calibration from issue cost and frequency, moments of the thresholds, and their Table 2 has the sign pattern of M3 in the delay and the fixed cost at small costs (`PeuraKeppo.table2_det_neg`). The parallel ends at the parameter, a delay rather than a preference, and at the method, a calibration rather than a local identification theorem. | "For D and K, we identify common estimates across banks such that their implied capital raising cost and frequency are of plausible magnitude." | 2184 | M3 |
| L9 | Dai et al. [2019] | Their Table 2.1 gives the responses of the dividend and recapitalisation barriers to seven parameters, the fixed cost with the signs of M1 and M2 (`DaiHuangKeppo.table21_K_signs`). They state no identification and have no shareholder asymmetry parameter, and their delay and fixed cost move the trigger the same way. | "K 0.0017 −0.0543 0 −0.0004" | 76 | M1, M2, M3 |
| L10 | Bolton et al. [2011] | Its authors state that the double barrier of payout and external financing is an established structure. Its fixed issuance cost is proportional to capital to keep the model homogeneous, as in M1 to M3, and each issue returns to one target, as in M7. | "While this double-barrier policy is not new (it goes back to the inventory model of Miller and Orr 1966 in corporate finance), our model provides substantial new insight" | 3 | M1, M7 |

| ID | Paper | What we rely on | Quote | Page | Rows |
|---|---|---|---|---|---|
| L11 | Løkka and Zervos [2005] | With proportional issuance costs only, issuance is a minimal reflection at zero, neither a lump nor a target. The fixed issuance cost is what produces the single target of M7. | "In this problem, it is optimal to issue minimal new equity so as to reflect the reserves process at 0 and make minimal dividend payments so as to reflect the reserves process at a certain level $x^*$ (see Section 4)." | 2 | M7 |
| L12 | Barberis et al. [2001] | With a constant level of loss aversion, the volatility of log returns equals that of dividend growth for every level, the parallel to the absence of $\lambda_S$ from the limits of M11. The parallel ends at the mechanism, a constant price-dividend ratio in an endowment economy rather than a binding regulatory cap. | "the volatility of log returns in this model is equal to the volatility of log dividend growth, namely 12 percent." | 47 | M11 |
| L13 | Engle and Siriwardane [2018] | They split observed equity volatility asymmetry into a mechanical-leverage share and a market-exposure share. The parallel ends at the object, a response of variance to the sign of news rather than limits of variance across regions of a state variable. | "Volatility asymmetry is mostly explained by exposure to the aggregate market, not a mechanical leverage effect." | 449 | M11 |
| L14 | Li et al. [2023] | In an unconstrained loss-averse control the large-wealth limits of the optimal controls depend on the preference parameters, the contrast to M11, where a binding regulatory cap fixes them. | "our asymptotic limits differ significantly from the ones in the Merton's problem, which now sensitively depends on the reference degree parameter" | 13 | M11 |
| L15 | Berger et al. [2008] | Buffers well above the minimum are the fact the shortfall penalty of A4 and M8 is to explain. The parallel ends at the method, a partial-adjustment regression without a model of the thresholds. | "Since the early 1990s, however, the largest U.S. banking organizations have maintained ratios well in excess of regulatory minimums." | 1 | M8 |
| L16 | Milne and Whalley [2001] | The buffer guards against the supervisor's intervention or the market's response below the minimum, the reading of the penalty in Appendix D. Their cost is an audit at random times, not a graduated running penalty. | "What matters instead is the possibility of future capitalisation falling below minimum required levels and inducing a costly regulatory intervention or market response." | 3 | M8 |

| ID | Paper | What we rely on | Quote | Page | Rows |
|---|---|---|---|---|---|
| L17 | Peura and Keppo [2006] | The optimising bank with a fixed issuance cost (M6) and liquidation at the minimum underpredicts the capital level, on their own calibration. The shortfall penalty of A4, whose form M8 carries, is the friction added. | "The model falls somewhat short of explaining the level of capital held by banks." | 2188 | M6, M8 |
| L18 | Peura and Keppo [2006] | The graduated penalty they call for is $\Lambda$ of A4 and M8, with the asymmetry parameter $\lambda_S$ setting how much it charges. The parallel ends at the call, since they do not add it. | "A model in which the penalty gradually increases as the capital ratio deteriorates would therefore be more realistic." | 2192 | M8 |
| L19 | Peura and Keppo [2006] | The question they leave, whose cost binds the bank's capital. The thresholds identify the asymmetry parameter under any reading and do not choose among them. | "Whether these omissions matter reduces to the issue of which constituency provides the binding constraint on bank capitalization." | 2192 | M3, M8 |
| L20 | Peura and Keppo [2006] | Their parameters beyond the return moments are imposed, not identified from the bank's policy. M3 identifies two of them from the thresholds. | "imposed common values for other parameters (mainly guided by high level economic intuition)" | 2192 | M3 |
| L21 | Dai et al. [2019] | Their parameters are fitted to equity values, not identified from the thresholds. | "are calibrated by fitting the partially observed value function" | 84 | M3 |
| L22 | Peura and Keppo [2006] | Issues are larger and rarer when the fixed cost is high, the widening trigger-to-target gap of M2 and the finite count of M6. | "The owners optimally issue new equity in larger quantities but less frequently when the cost of issuance is high" | 2179 | M2, M6 |
| L23 | Peura and Keppo [2006] | In their model the fixed cost alone puts the issue barrier at the minimum, and the delay moves it up. In M2 the trigger is an interior threshold whose responses to the fixed issuance cost and to the asymmetry parameter are derived. | "The sole presence of the fixed cost does not generate a positive capital issue barrier and will not result in a positive probability of liquidation." | 2177 | M2 |
| L24 | Peura and Keppo [2006] | The penalty at the minimum is liquidation. The shortfall penalty of M8 grades it below the reference. | "We assume that the corrective action from violation of the minimum capital requirement will be liquidation" | 2168 | M8 |

| ID | Paper | What we rely on | Quote | Page | Rows |
|---|---|---|---|---|---|
| L25 | Dai et al. [2019] | The asymmetry in their model is the regulator's, between the costs of liquidating and of not liquidating, and it acts through the liquidation barrier. The asymmetry parameter of M2 and M3 is in the bank's objective and moves the trigger against the fixed issuance cost. | "when the liquidation parameter a rises, then the liquidation barrier I falls" | 71 | M2, M3 |
| L26 | Bayraktar et al. [2026] | The minimum of two branches with a switching point, the cap and the crossing point of M10. | "The key structure is the minimum operator in the regulatory constraint, with a switching point" | 15 | M10 |
| L27 | Bayraktar et al. [2026] | At zero equity the cap forces zero exposure and the diffusion vanishes, the boundary that M9 sends to minus infinity and near which M11 takes its deficit limit. | "At the hypothetical zero-equity level y = 1, the solvency constraint forces $\pi = 0$ and the diffusion coefficient of the ratio process vanishes." | 11 | M9, M11 |
| L28 | Bayraktar et al. [2026] | Their dividend barrier and post-issue target are the paper's. Their trigger is the exogenous distress boundary, where the trigger of M2 is interior and responds to the two parameters. | "dividends are paid at an upper barrier, while recapitalization, when optimal, is triggered at the distress boundary and jumps to a post-issuance target" | 12 | M1, M2, M7 |
| L29 | Bayraktar et al. [2026] | The cap moves the thresholds and does not create them, as in M1 and M2, where the costs make the thresholds and the cap of M10 shapes the variance. | "The regulatory constraints do not create the threshold structure from scratch" | 13 | M1, M2, M10 |
| L30 | Bayraktar et al. [2026] | The liability shock is correlated with the risky return, as in A1, and enters the variance of M10 to M12. | "c is their shock correlation" | 13 | M10, M11, M12 |
| L31 | Bayraktar et al. [2026] | Their parameters are illustrative and none is read off the policy, where M3 reads two off the thresholds. | "These values constitute an illustrative parameterization rather than an empirical calibration" | 13 | M3 |
| L32 | Black and Perold [1992] | Their safe asset can track the liability, so the liability is hedged. In M10 to M12 the liability shock enters buffer variance unhedged. | "'safe' may mean that it closely tracks a liability stream" | 404 | M10, M11 |
| L33 | Barberis et al. [2001] | Volatility from loss aversion needs loss aversion to vary over time. In M11 a constant asymmetry parameter leaves no mark on the limits of buffer variance, for a different reason, the binding cap. | "The investor's risk aversion is then constant over time, and stock prices lose an important source of volatility." | 4 | M11 |

| ID | Paper | What we rely on | Quote | Page | Rows |
|---|---|---|---|---|---|
| L34 | Engle and Siriwardane [2018] | Equity variance is the squared leverage multiplier times asset variance, the structural relation behind the change of variable of M10. | “which is the exact relationship implied by structural models of credit” | 459 | M10 |
| L35 | Bayraktar et al. [2026] | Without borrowing the liquidity branch does not bind in the range of parameters they solve. C3 gives the condition under which it does not bind at any state. | “the LCR constraint is nonbinding within the parameter range considered” | 20 | M11, M12 |
| L36 | Barberis et al. [2001] | Their loss-averse preference is kinked-linear at the reference, the form of the shortfall penalty of M8. It values gains and losses on an asset, not a shortfall of a bank’s capital below its requirement. | “This is a piecewise linear function, shown as the solid line in Figure I. It is kinked at the origin, where the gain equals zero.” | 10 | M8 |
| L37 | Milne and Whalley [2001] | The cost of falling below the minimum, through the audit, and the fixed cost of issue are joint determinants of the buffer, the two frictions that the asymmetry parameter and the fixed issuance cost stand for in A4 and A5. | “desired bank capital holdings are a function of cash-flow uncertainty, franchise value, audit frequency, recapitalisation costs, and cost of equity relative to deposits” | 9 | M2, M3 |
| L38 | Milne and Whalley [2001] | Both frictions in one bank make the capital decision an inventory problem with a target, the counterpart of the injection target of M7. Its issue point is not a choice, since the bank issues only when an audit finds it below the minimum (L64), where the recapitalisation trigger of M2 is chosen. | “The bank capital decision is then akin to an (s, S) inventory problem.” | 9 | M2, M7 |
| L39 | Milne and Whalley [2001] | Result 12. A higher fixed cost of issue or more frequent audit raises desired capital. | “always results in an increase in desired capital” | 23 | M2, M3 |
| L40 | Milne and Whalley [2001] | Result 12. More frequent audit moves the policy the same way as a higher fixed cost of issue, so the two frictions have like responses, the case in which the control of M3 shows the Jacobian can be singular. | “A shift in the same direction would be produced by an increase of q.” | 23 | M3 |

| ID | Paper | What we rely on | Quote | Page | Rows |
|---|---|---|---|---|---|
| L41 | Peura and Keppo [2006] | At $(\Delta, K) = (0.5, 1.00)$ the printed values, with L42 and L43, give a determinant of the dividend and issue barriers in the delay and the fixed cost of $+1/25$ (`PeuraKeppo.table2_det_high_K`), against $-29/100$ at small costs (`PeuraKeppo.table2_det_value`). The values are printed to one decimal. | ".5 1.00 .9 3.7 2.8 27 122" | 2184 | M3 |
| L42 | Peura and Keppo [2006] | At the highest cost a longer delay lowers the issue barrier, so the delay and the fixed cost move it the same way there. | "1 1.00 .6 3.8 3.2 24 291" | 2184 | M3 |
| L43 | Peura and Keppo [2006] | The cost step of the determinant at the highest cost. | ".5 .50 1.1 3.5 2.4 17 57" | 2184 | M3 |
| L44 | Dai et al. [2019] | With L9, the elasticity determinant of the two barriers in the delay and the fixed cost is $-103549/25000000$, about $-0.0041$ (`DaiHuangKeppo.table21_det_value`), non-zero by magnitude while the two move the issue barrier the same way. | "Δ 0.0808 −0.1444 0 −0.0146" | 76 | M3 |
| L45 | Bayraktar et al. [2026] | The cap geometry of the baseline at which M13 evaluates the limits of buffer variance. | "The baseline regulatory parameters are (a1 , a2 , a3 ) = (0.045, 0.05, 0.30)" | 13 | M13 |
| L46 | Bayraktar et al. [2026] | The asset volatility, the liability volatility and the return-liability correlation of the baseline of M13. | "r = 0.02, $\mu$ = 0.04, $\mu$L = 0.03, $\rho$ = 0.12, $\gamma$ = 0.02, together with $\sigma$ = 0.08, $\sigma$L = 0.03, c = 0.20, $\kappa$ = 0.01, $\kappa'$ = 0.02" | 13 | M13 |
| L47 | Dai et al. [2019] | Their bank has no liability shock, which in M10 to M12 enters the limits of buffer variance. | "We ignore debt volatility and debt financing" | 26 | M10, M11 |
| L48 | Constantinides and Richard [1978] | A running penalty that grows linearly below a level and a fixed and a proportional cost of each transfer in one objective, the pairing of A4 and A5 in cash management rather than in a bank. Their trigger and target below the level are the recapitalisation trigger and the injection target of M2 and M7. | "model with both fixed and proportional transactions costs and with linear holding and penalty costs" | 620 | M2, M7, M8 |

| ID | Paper | What we rely on | Quote | Page | Rows |
|---|---|---|---|---|---|
| L49 | Constantinides and Richard [1978] | Their comparative statics are confined to special cases, an approximation with no drift and vanishing discounting, where M1 and M2 hold under the hypotheses of Appendix B. | "In this section and under additional assumptions we explicitly solve for the policy parameters and perform a comparative statics analysis." | 630 | M1, M2 |
| L50 | Constantinides and Richard [1978] | In that approximation the upper trigger rises in the penalty and in the fixed cost of raising cash, the signs of M1 for the dividend barrier. They state no direction for the lower trigger or the lower target. | "the policy parameter u is an increasing function" | 631 | M1 |
| L51 | Lang and Menno [2026] | Their cost of breaching the buffer requirement is a charge per unit of loans that is the same at every shortfall (`LangMenno.stigma_flat_below`), and with it the bank holds capital exactly at the buffer requirement. The shortfall penalty of M8 grows with the shortfall (`LangMenno.graduated_grows_below`). | "Banks do not maintain voluntary capital buffers above the CBR" | 18 | M8 |
| L52 | Constantinides and Richard [1978] | In their approximation the four thresholds pin the penalty and the fixed cost of raising cash once the holding cost is known (`Constantinides.thresholds_pin_pK`), an identification the authors do not state. M3 identifies the asymmetry parameter and the fixed issuance cost from two thresholds without a holding cost. | "We next consider the case where costs are not symmetric and there are fixed as well as proportional transactions costs." | 631 | M3 |
| L53 | Behn et al. [2019] | Their cost of breaching the capital requirement is a charge that does not grow with the shortfall (`BehnEtAl.breach_flat_below`), the step the shortfall penalty of M8 grades. | "These costs are reminiscent of the bankruptcy cost in classical trade-off theory models" | 10 | M8 |
| L54 | Behn et al. [2019] | They estimate the breach charge and an issuance cost with no fixed part (`BehnEtAl.issueCost_no_fixed_part`) in one bank, and separate the two by the moments each is matched to rather than by the thresholds of the policy, as M3 does. | "The costs of breaching the minimum capital and liquidity requirements are mainly identified by the extent of the observed voluntary buffers" | 23 | M3 |

| ID | Paper | What we rely on | Quote | Page | Rows |
|---|---|---|---|---|---|
| L55 | Glover et al. [2022] | A breach forces an issue priced at the proportional issuance cost, so one parameter charges both the shortfall and the issue (`GloverEtAl.breach_slope_is_issuance_cost`) and no policy can separate them. M3 has two parameters, the asymmetry parameter and the fixed issuance cost. | "whenever the bank decides (or is forced) to raise new equity it will incur a cost" | 10 | M3, M8 |
| L56 | Lambertini and Uysal [2014] | The capital requirement is a smooth log penalty on net worth, negative below required capital and a reward above it (`LambertiniUysal.logTerm_pos_above`), with no kink. The shortfall penalty of M8 is zero at and above the reference and kinked there. | "Due to the log term, the bank is penalized when its capital goes below the required 8 percent of total assets" | 23 | M8 |
| L57 | Décamps et al. [2011] | A fixed cost of each issue makes issues lumpy (`DecampsEtAl.unitCost_falls`), the single target of M7. | "Because issuing shares involves a fixed cost, equity adjustments take place in lumpy and infrequent issues" | 1503 | M7 |
| L58 | Alvarez et al. [2014] | In a fixed-cost inaction model the parameters are identified globally from moments of the distribution of adjustments rather than from thresholds, and two parameters that both raise kurtosis are separated only by adding the observed cost of adjustment (`AlvarezLeBihanLippi.control_same_direction_not_identified`). M3 identifies two parameters from two thresholds by the opposite signs of their effects on one of them. | "the kurtosis of the price changes would not change. Moreover, kurtosis is increasing in both n and" | 24 | M3 |
| L59 | Federico et al. [2026] | At required capital the bank issues at a fixed cost or is liquidated, and reserves never go below it (`FedericoEtAl.liquidationPayoff_nonneg`), so no cost grows with a shortfall. The shortfall penalty of M8 grades the cost below the reference. | "When this occurs, the manager must either liquidate the bank or recapitalize it by issuing new equity at a cost." | 10 | M8 |

| ID | Paper | What we rely on | Quote | Page | Rows |
|---|---|---|---|---|---|
| L60 | Federico et al. [2026] | The issue trigger is required capital itself, where the trigger of M2 is interior and responds to the asymmetry parameter and the fixed issuance cost. Their dividend restriction is a payout tax that depends on the macro state, not on the shortfall, and the responses of their thresholds are numerical at one calibration. | “the bank decides whether to default or, at a cost, to issue equity (and in what amount) when capital requirements become binding” | 3 | M2 |
| L61 | Bonaparte et al. [2011] | Preference parameters and a fixed adjustment cost are estimated together, and the table of elasticities of the moments to the parameters is their only device for identification. The thresholds of their inaction band are not used, where M3 identifies from two thresholds. | “This table is also informative about identification as it indicates which moments are particularly sensitive to variations in each of the parameters.” | 8 | M3 |
| L62 | Milne and Whalley [2001] | Result 6. With no return to bearing risk the bank takes the smaller of two assumed volatilities where its value is concave and the larger where it is convex. The exposure rule of the cap term reduces to theirs when the risk premium is zero, with no exposure for the smaller and the regulatory cap for the larger, so the bound on risk is set by regulation here and assumed there. | “optimal policy is always to extremise the degree of cash flow uncertainty” | 18 | M11, M12 |
| L63 | Milne and Whalley [2001] | Result 5. Below the minimum, which the bank can reach because the audit is random, its value is convex and it takes the most risk. The value function of M4 is convex just above the recapitalisation trigger instead, where the bank issues rather than waits for an audit. | “a bank with less than the minimum required capital and hence under immediate threat of regulatory intervention is risk loving” | 18 | M4 |
| L64 | Milne and Whalley [2001] | Result 8. The bank issues or liquidates only after an audit finds it below the minimum and never by choice, so their model has no recapitalisation trigger in the sense of M2. | “A bank never chooses voluntary recapitalisation or voluntary liquidation.” | 19 | M2 |

| ID | Paper | What we rely on | Quote | Page | Rows |
|---|---|---|---|---|---|
| L65 | Milne and Whalley [2001] | The audit arrives at a constant rate, so below the minimum their equation charges at that rate the value lost by being forced to issue or close, which depends on the value function. With a fine in proportion to the shortfall in place of forced action, the same rate gives the shortfall penalty of A4, with the asymmetry parameter less one equal to the rate times the fine (Section 4.3). | "that regulators engage in a Poisson audit" | 9 | M8 |
| L66 | Milne and Whalley [2001] | The bank literature writes the objective as linear in the bank's cash flows or as a utility concave in them, the second justified by a divergence of interest between managers and shareholders. They take the first, as does the paper with the shortfall penalty added (Appendix D). | "The alternative to assuming a linear bank objective is a risk-averse bank utility function" | 11 | M8 |
| L67 | Milne and Whalley [2001] | By their account of two papers the paper has not read, in portfolio models with a concave utility a capital requirement can shift the bank towards riskier assets. Whether a higher solvency floor raises the exposure the bank chooses below the cap is a question of the comparative statics (Section 3). | "that the introduction of regulatory capital requirements may induce banks to shift their portfolio composition towards riskier assets" | 11 | M12 |
| L68 | Milne and Whalley [2001] | With a linear objective the frictions induce the bank's attitude to risk. In the paper's model the value function is concave at the injection target, a hypothesis of M2, and convex just above the recapitalisation trigger (M4), so the bank is averse to risk at the first and not at the second. | "we do find induced risk-aversion when capital is less than the desired level" | 11 | M2, M4 |

| ID | Paper | What we rely on | Quote | Page | Rows |
|---|---|---|---|---|---|
| L69 | Milne and Whalley [2001] | Result 13. A higher minimum raises their desired capital one for one and leaves the buffer above the minimum unchanged, by the homogeneity of their equation in capital less the minimum. In the paper's model required capital enters the shortfall penalty and not the cap, so the homogeneity fails, and in the solve the dividend barrier rises by about three-eighths of a rise in required capital and the recapitalisation trigger by about a seventh (Section 3). | "an alteration of the minimum capital requirement has no long run impact on either the buffer of free capital" | 23 | M8 |
| L70 | Engle and Siriwardane [2018] | Their sample is financial firms, and they use the structural relation to back out asset returns and not asset levels. It bears on buffer variance, the object of M10 and M15, and not on the thresholds, which are levels of the state. | "we intentionally avoid taking a stance on a specific asset pricing model, and why we will only use our model to back out asset returns, not asset levels" | 460 | M10, M15 |
| L71 | Calem and Rob [1999] | In a dynamic bank that rebuilds capital only from its earnings and cannot issue, a severely undercapitalized bank takes the most risk it can. The paper's bank issues at the recapitalisation trigger, and there the regulatory cap bounds its exposure, so the convexity of M4 does not become maximal risk. | "In general, a severely undercapitalized bank takes maximal risk in an effort to improve its capital position." | 18 | M4 |
| L72 | Calem and Rob [1999] | A higher flat requirement raises the risk a well-capitalized bank takes, the dynamic counterpart of the portfolio result of L67. At the paper's baseline a higher solvency floor lowers the exposure through the cap (Section 3.5). | "the bank generally takes on more portfolio risk" | 23 | M12 |
| L73 | Calem and Rob [1999] | A premium charged below required capital, the same at every shortfall, raises the risk undercapitalized banks take because it slows recapitalisation from earnings. In the paper's model the shortfall penalty grows with the shortfall and the bank can issue, so the asymmetry parameter raises the recapitalisation trigger (M2) rather than the risk taken. | "The underlying intuition is that the premium surcharge cuts into bank earnings, hampering the effort to recapitalize." | 25 | M2, M8 |

| ID | Paper | What we rely on | Quote | Page | Rows |
|---|---|---|---|---|---|
| L74 | Calem and Rob [1999] | Their risk-based standard bounds the risky share linearly in capital (`CalemRob.riskbased_is_linear_bound`), the form of the solvency branch of the regulatory cap with an intercept, and it moderates risk only when stringent enough. | "be sufficiently stringent for it to be effective" | 34 | M10 |
| L75 | Fries et al. [1997] | Equity holders inject capital to keep the bank open and stop when injecting no longer pays, so the bank closes at a point they choose, below zero net assets (`FriesEtAl.closure_factor_in_unit`). The paper's bank always recapitalises at the trigger and has no option to abandon, a choice of scope (Section 2.3). | "the willingness of equity-holders to inject capital constrains banking policy" | 444 | M2 |
| L76 | Fries et al. [1997] | Their injections are continuous, a payout to shareholders that turns negative, with no fixed cost. With the fixed issuance cost of A5 injections are lumpy and finite in number (M6) and land on one target (M7). | "we allow cash disbursements to shareholders to become negative in some states of the world" | 446 | M6, M7 |
| L77 | Fries et al. [1997] | Limited liability makes equity convex near a late closure, and they call a value that would rise with volatility moral hazard. The value function of M4 is convex just above the recapitalisation trigger because of the option to issue rather than limited liability, and there the regulatory cap bounds the exposure. | "Late closure creates an option value for equity-holders which makes their claims convex in the state-variable." | 460 | M4 |
| L78 | Fries et al. [1997] | Values per unit of deposits, by the homogeneity of the equation and its boundary conditions, the reduction the paper keeps by charging the fixed issuance cost per unit of liabilities (L4). | "one may exploit the homogeneity of the differential equation and the boundary conditions" | 448 | M1, M2, M3 |
| L79 | Engle and Siriwardane [2018] | Their leverage multiplier is a function of observed leverage. In M14 and M15 the multiplier is the risky exposure over equity, which the bank chooses and the regulatory cap fixes where it binds. | "the leverage multiplier becomes a simple function of observable leverage" | 450 | M14, M15 |

| ID | Paper | What we rely on | Quote | Page | Rows |
|---|---|---|---|---|---|
| L80 | Engle and Siriwardane [2018] | Their multiplier rises in leverage, so equity volatility rises as equity falls for a given asset volatility. Where the cap binds the paper's multiplier is constant on each branch, so within a branch buffer variance has no such effect (M10, M11). | "A second common feature is that the leverage multiplier is monotonically increasing in leverage." | 456 | M10, M11 |
| L81 | Milne and Robertson [1996] | A firm that cannot issue pays dividends at a barrier where $V' = 1$ and $V'' = 0$, the conditions M1 uses at the dividend barrier, and is liquidated at a threshold. It is the base model of Peura and Keppo [2006] and Milne and Whalley [2001] (L7, L37). | "This implies that the firm cannot raise new equity finance." | 6 | M1 |
| L82 | Milne and Robertson [1996] | They offer a penalty rate on debt below the threshold in place of liquidation, and do not solve that model. The shortfall penalty of A4 is a running charge of that kind, graded below required capital (M8). | "This penalty interest rate would induce a similar aversion to indebtedness as appears in our model." | 7 | M8 |
| L83 | Milne and Robertson [1996] | With liquidation and no option to issue the value function is concave below the barrier and aversion to risk rises as cash falls, so convexity appears only past the liquidation threshold. The value function of M4 is convex just above the recapitalisation trigger because the bank issues there. | "the degree of local risk aversion increases as the firm moves closer to liquidation" | 15 | M4 |

Every paper the tables name has been read in full and has a record with Lean under `theory/lit/`.

## 4.5 Papers cited

Table 11 lists every paper the manuscript cites, with its title, the version read, where it is published, the number of citations and the rating of the journal in the 2025 ABDC Journal Quality List. The column Claims counts the rows of Table 10 that rest on the paper and links to the first of them, and the last column says in the names of the measurement map what the manuscript loses if the paper is not relied on. The papers are in descending order of claims, papers with as many claims are in descending order of citations, counting the other versions, and papers equal on both are in order of bib key. Citations are the count of the OpenAlex record of the published version, or of the working paper where none was found, retrieved on 30 October 2026, with the counts of OpenAlex's records of other versions of the same paper in brackets. The last column links to each of these records, the one counted first. OpenAlex counts are usually lower than those of Google Scholar. The date is that of the printed issue registered with Crossref, or of the working paper. Where the published version differs from the version read, the published version was not read.

Table 11: Papers cited, in order of claims and then citations, with the title, the version read, the published version, its journal's rating, the records counted, the rows of Table 10 that rest on it, and what the manuscript loses without it.

| Paper | Title | Version read | Published as | Date | Citations | Claims | ABDC 2025 | OpenAlex | If not relied on |
|---|---|---|---|---|---|---|---|---|---|
| Milne and Whalley [2001] | Bank Capital Regulation and Incentives for Risk-Taking | City University working paper | working paper (SSRN 299319) | 2001-12 | 147 | 13 | – | main | A cost below required capital paired with a fixed issuance cost in one bank loses its earliest statement, and the case of like signs, where the determinant of M3 can vanish, loses its printed example. |
| Bayraktar et al. [2026] | Tractable Bank Capital Structure: Optimal Control under Basel III Constraints | arXiv 2603.14557v2 | working paper (arXiv, SSRN) | 2026-03 | 0 | 13 | – | main, other 1, other 2 | The regulatory cap, its crossing point and the baseline of M13 lose their source, and C3 and the contrast with an exogenous recapitalisation trigger go. |
| Peura and Keppo [2006] | Optimal Bank Capital with Costly Recapitalization | the article | J. Business 79(4), 2163–2201 | 2006-07 | 188 | 12 | not listed (A* in 2016, discontinued) | main | The fixed issuance cost and the single injection target lose their bank precedent, the shortfall penalty is no longer a call in print, and C4 and C6 lose their main comparison. |
| Engle and Siriwardane [2018] | Structural GARCH: The Volatility-Leverage Connection | the article | RFS 31(2), 449–492 | 2018-02 | 30 (+18) | 5 | A* | main, other 1, other 2 | The change of variable from asset volatility to buffer variance loses its empirical counterpart, and the contrast with the sign asymmetry of equity volatility goes. |
| Dai et al. [2019] | Opaque Bank Assets and Optimal Equity Capital | Huang's thesis, ch. 2, arXiv 2107.02242v1 | JEDC 100, 369–394 | 2019-03 | 11 | 5 | A* | main | C4 loses one of its two numerical precedents, and the claim that issuance-cost models fit rather than identify their parameters rests on Peura and Keppo [2006] alone. |
| Calem and Rob [1999] | The Impact of Capital-Based Regulation on Bank Risk-Taking | FEDS 96-12 (1996) | J. Financ. Intermed. 8(4), 317–352 | 1999-10 | 410 (+74) | 4 | A* | main, other 1, other 2 | The contrast of a bank that gambles near insolvency because it cannot issue, and of a requirement that raises risk, with a bank that issues at the trigger under a cap goes, and Section 3.5 loses its dynamic precedent. |
| Constantinides and Richard [1978] | Existence of Optimal Simple Policies for Discounted-Cost Inventory and Cash Management in Continuous Time | the article | Oper. Res. 26(4), 620–636 | 1978-08 | 186 | 4 | A* | main | The closest precedent outside banking for a linear shortfall penalty with a fixed issuance cost goes, and C4 no longer notes thresholds that pin both parameters there. |

| Paper | Title | Version read | Published as | Date | Citations | Claims | ABDC 2025 | OpenAlex | If not relied on |
|---|---|---|---|---|---|---|---|---|---|
| Fries et al. [1997] | Optimal Bank Reorganization and the Fair Pricing of Deposit Guarantees | the article | J. Bank. Financ. 21(4), 441–468 | 1997-04 | 62 | 4 | A* | main | Closure chosen by shareholders who stop injecting capital, the alternative to a bank that always recapitalises, loses its source, and continuous injections lose their contrast with M6 and M7. |
| Barberis et al. [2001] | Prospect Theory and Asset Prices | the article | QJE 116(1), 1–53 | 2001-02 | 1877 (+244) | 3 | A* | main, other 1, other 2 | The kinked form of the shortfall penalty loses its source, and the contrast of K2 with an asymmetry read off the volatility of returns loses its main case. |
| Black and Perold [1992] | Theory of Constant Proportion Portfolio Insurance | the article | JEDC 16(3–4), 403–426 | 1992-07 | 403 | 3 | A* | main | The constant buffer variance on the solvency branch is no longer tied to capped portfolio insurance, and C2 loses the case where the same limits carry preferences. |
| Milne and Robertson [1996] | Firm Behaviour under the Threat of Liquidation | draft of February 1994 | JEDC 20(8), 1427–1449 | 1996-08 | 73 (+1) | 3 | A* | main, other 1 | The bank papers lose their no-issuance base, the penalty rate offered in place of liquidation loses its source, and the contrast of concavity without an option to issue with the convexity of M4 goes. |
| Behn et al. [2019] | A Dynamic Model of Bank Behaviour under Multiple Regulatory Constraints | ECB WP 2233 | working paper | 2019-01 | 14 | 2 | – | main | The flat charge for breaching required capital, which the shortfall penalty grades, rests on Lang and Menno [2026] alone, and separating two costs by moments rather than thresholds loses its bank example. |
| Federico et al. [2026] | Coordinating Bank Dividend and Capital Regulation | SAFE WP 487 | Math. Finance, online, no issue yet | 2026-09 | 0 | 2 | A | main, other 1, other 2 | A recapitalisation trigger fixed at required capital loses its recent model, and the contrast with the interior trigger of M2 rests on Peura and Keppo [2006]. |
| Bolton et al. [2011] | A Unified Theory of Tobin's q, Corporate Investment, Financing, and Risk Management | accepted manuscript | J. Finance 66(5), 1545–1578 | 2011-10 | 663 (+266) | 1 | A* | main, other 1, other 2 | The double barrier of dividends and issuance is no longer shown to be established, and a fixed issuance cost scaled to keep one dimension loses a precedent. |
| Berger et al. [2008] | How Do Large Banking Organizations Manage Their Capital Ratios? | the article | J. Financial Serv. Res. 34(2–3), 123–149 | 2008-12 | 531 (+11) | 1 | A | main, other 1 | The fact that banks hold buffers well above required capital, which the shortfall penalty is to explain, loses its empirical source. |

| Paper | Title | Version read | Published as | Date | Citations | Claims | ABDC 2025 | OpenAlex | If not relied on |
|---|---|---|---|---|---|---|---|---|---|
| Alvarez et al. [2014] | Small and Large Price Changes and the Propagation of Monetary Shocks | NBER WP 20155 | Amer. Econ. Rev. 106(10), 2817–2851, as "The Real Effects of Monetary Shocks in Sticky Price Models" | 2016-10 | 169 (+56) | 1 | A* | main, other 1, other 2, other 3, other 4 | C4 loses its comparison with identification from moments of adjustments rather than from thresholds. |
| Décamps et al. [2011] | Free Cash Flow, Issuance Costs, and Stock Prices | the article | J. Finance 66(5), 1501–1544 | 2011-10 | 166 (+12) | 1 | A* | main, other 1 | The lumpy issues of M7 lose one precedent and keep Peura and Keppo [2006] and Bolton et al. [2011]. |
| Løkka and Zervos [2005] | Optimal Dividend and Issuance of Equity Policies in the Presence of Proportional Costs | preprint of July 2005 | Insurance Math. Econ. 42(3), 954–961 | 2008-06 | 117 | 1 | A* | main | Nothing read shows that proportional issuance costs alone give no injection target, so M7 loses its contrast. |
| Bonaparte et al. [2011] | Consumption Smoothing and Portfolio Rebalancing: The Effects of Adjustment Costs | NBER WP 16957 | J. Monetary Econ. 59(8), 751–768 | 2012-12 | 29 | 1 | A* | main | C4 loses its comparison with joint estimation of a preference and a fixed cost that leaves the inaction thresholds unused. |
| Li et al. [2023] | Optimal Consumption with Loss Aversion and Reference to Past Spending Maximum | arXiv 2108.02648v4 | SIAM J. Financial Math. 15(1), 121–160 | 2024-03 | 12 | 1 | B | main | K2 loses its contrast with limits of controls that carry a preference parameter when no cap binds. |
| Lambertini and Uysal [2014] | Macroeconomic Implications of Bank Capital Requirements | EPFL working paper | working paper | 2014-02 | 5 | 1 | – | main | The smooth penalty on net worth, the alternative to the kinked shortfall penalty, loses its source. |
| Glover et al. [2022] | Capital Ideas: Optimal Capital Accumulation Strategies for a Bank and its Regulator | preprint of 1 December 2022 | European J. Finance 29(18), 2075–2106 | 2023-12 | 0 | 1 | A | main | M3 loses its contrast with a single parameter that prices both the shortfall and the issue, which no policy can separate. |
| Lang and Menno [2026] | A Structural Model of Capital Buffer Usability | ECB WP 3188 | working paper | 2026 | 0 | 1 | – | main, other 1 | The flat charge for breaching required capital, which the shortfall penalty grades, rests on Behn et al. [2019] alone. |

# 5 What the paper adds to the three approaches

This section sets each theoretical result of the paper against the three approaches of Section 4.2, and says what the result adds that none of them has and what it shares with them. The comparison is with the papers read in full, rather than with the literature at large. The claim that K1 is new is concluded against those papers (C4), and the claim that K2 is new waits on what Table 13 names (C5).

The identification of K1 adds a second friction to the issuance-cost models and a theorem to their tables. The responses of the dividend barrier and the recapitalisation trigger to the fixed issuance cost are tabulated there (L9, L22), and in the table of Peura and Keppo [2006] the issue delay and the fixed issuance cost move the trigger in opposite directions at small costs, with both set by calibration to issue cost and frequency (L8). The paper's second parameter is the asymmetry parameter of the graduated shortfall penalty that Section 4 traces (L18, L24), rather than a delay. The responses of the two thresholds to the asymmetry parameter and to the fixed issuance cost are derived under the hypotheses of Appendix B (M1, M2) rather than read off a solved case, and the opposite effect of the two parameters on the trigger gives their local identification from the two thresholds (M3), where the papers read set their parameters (L8, L20, L21). Regulatory-cap control has an exogenous trigger and no asymmetry parameter (L28), and the preference approach has no trigger. K1 would be in print if a paper read derived the responses of the thresholds to the asymmetry parameter of a shortfall penalty in a bank's objective, and none of the three approaches has such a parameter in a bank's objective (L12, L24, L28).

The limits of buffer variance in K2 add what regulatory-cap control leaves out. The regulatory cap with its two branches is that of Bayraktar et al. [2026] (L1, L26), and the constant buffer variance on the solvency branch is the constant log-cushion volatility of capped portfolio insurance (L5), so neither is claimed new. The deficit limit and the surplus limit where the exposure binds the cap, and the bound for every exposure within it (M11, M12), are not evaluated by Bayraktar et al. [2026] (L1). That the limits do not carry the asymmetry parameter (M11) sets them apart from the two papers read in which limits of a volatility carry a preference (L6, L14), as Section 4 sets out, and the liability shock enters them unhedged, where the safe asset of Black and Perold [1992] can hedge it (L32). K2 would be in print if a paper read evaluated the limits of buffer variance under a regulatory cap with an unhedged liability shock, and none of the three approaches does. That the optimal exposure binds the cap near both ends is a hypothesis of M11, and it is part of what C5 waits on.

K1 and K2 together place the asymmetry parameter in the thresholds and not in the limits of buffer variance, on one bank. The preference approach looks for the mark of an asymmetry in the variance of returns, and finds none when the asymmetry is constant (L12), while the issuance-cost models have thresholds and no asymmetry parameter in the bank's objective (L24). Under the paper's regulatory cap the limits of buffer variance carry nothing about the asymmetry parameter (M11) and the two thresholds identify it (M3), so the paper reads the asymmetry parameter off the thresholds, where the preference approach reads an asymmetry off the variance of returns.

C3 replaces a range of parameters by a condition. Bayraktar et al. [2026] find that without borrowing the liquidity branch does not bind in the range they solve (L2, L35), and C3 states that it does not bind at any state where $a_2 \leq (1 - a_3)x$, so that the surplus limit of M11 belongs to the leveraged case.

The rest of the rows are shared with the issuance-cost models or are steps of the argument, and the paper does not claim them new. The pair of a dividend barrier and a recapitalisation trigger is an established double barrier (L10), each issue returns to one target as in Peura and Keppo [2006] and Bolton et al. [2011] (L7, L10), which is M7, and a proportional cost alone gives no target (L11). The kinked-linear form of the shortfall penalty is that of the loss-averse preference of Barberis et al. [2001] (L36), there over gains and losses on an asset and here over a shortfall of equity below required capital (M8). The absence of a kink at the recapitalisation trigger and the convexity just above it (M5, M4), the finite number of issues (M6) and the capital buffer as a coordinate (M9) are steps that the derivations of M1 to M3 and M10 to M12 use.

# 6 Conclusions

Table 12 holds the conclusions that rest on model rows and on papers read in full. K1 rests on M1 to M4 and gives C1, C4 and C6. K2 rests on M3 and M10 to M12 and gives C2 and C3, with C5 pending. Table 13 holds every claim still waiting, and each of its rows names what it waits on.

Table 12: Conclusions resting on model rows and papers read.

| ID | Statement | Rows |
|---|---|---|
| C1 | Under the hypotheses of Appendix B, the asymmetry parameter $\lambda_S$ and the fixed issuance cost $K$ are locally identified from the dividend barrier and the recapitalisation trigger. Both parameters raise the barrier, and they move the trigger in opposite directions, so the Jacobian of the two thresholds is non-singular. | M1, M2, M3, M4 |
| C2 | Wherever the optimal exposure binds the cap near both ends, the limits of buffer variance are $v_1^2$ and $v_3^2$, functions of $(\sigma, \sigma_L, c, a_1, a_3)$ only, and for every compliant exposure buffer variance is bounded by $\max\{\sigma_L^2, v_1^2\}$. The asymmetry parameter moves the thresholds (C1) and is absent from these limits and bounds. In capped portfolio insurance, whose geometry the cap shares, the slopes come from preferences, so the same limits would carry them. | M3, M10, M11, M12, L5, L6 |
| C3 | The surplus limit $v_3^2$ requires the liquidity branch to bind deep in surplus, which requires borrowing to hold the risky asset. With no borrowing, the main banking case of Bayraktar et al. [2026], the liquidity branch never binds once $a_2 \leq (1 - a_3)x$, so the deficit limit and the bound of M12 are the results that apply there. | M11, M12, L2 |
| C4 | In a bank's objective, the joint local identification of C1, of a shareholder asymmetry parameter of a shortfall penalty and a fixed issuance cost from the dividend barrier and the recapitalisation trigger by the opposite signs of their effects on the trigger, is new against the papers read in full. Its numerical precedents in other parameters are Peura and Keppo [2006] and Dai et al. [2019] (L8, L9). Outside banking, in the approximation of Constantinides and Richard [1978], four thresholds pin a linear penalty and a fixed cost once a holding cost is known, a step the authors do not take (L52), and Alvarez et al. [2014] identify a fixed-cost inaction model from moments of the distribution of adjustments rather than from thresholds, where two parameters that move one moment the same way need a second moment (L58). Bonaparte et al. [2011] estimate preference parameters and a fixed adjustment cost together, and their only device for identification is a table of the elasticities of their moments, with the thresholds of their inaction band not used (L61). | M3, L8, L9, L52, L58, L61 |
| C6 | C1 supplies what the specification Peura and Keppo [2006] call for (L18) lacks in their calibration. The asymmetry parameter and the fixed issuance cost are set by the two thresholds (M3) rather than imposed (L20) or fitted to equity values (L21). The thresholds set them in a model without the delay on issues of Peura and Keppo [2006], which the thresholds would read as a higher asymmetry parameter (L23, Section 3.3). Which constituency the shortfall penalty stands for (L19) is not settled by the thresholds (Appendix D). | M3, L18, L19, L20, L21, L23 |

Table 13: Claims not yet verified, each naming what it waits on.

| ID | Statement | Rows |
|---|---|---|
| C5 | That under a regulatory two-branch cap with an unhedged liability shock the limits of buffer variance carry no information about the asymmetry parameter, while the thresholds do (C2), is new. *Pending:* Choulli, Taksar and Zhou (2003), SIAM J. Control Optim. 41(6), read in full with a record; and a proof that the optimal exposure binds the cap near both ends, the hypothesis of M11. | M11, L5, L6 |

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# A Proofs

**Notation.** The state is the assets-to-liabilities ratio $x \in (1,\infty)$, $q > 0$ required capital as a ratio, and $z(x) = \log\big((x-1)/q\big)$ the capital buffer. The risky exposure is $\pi(x)$, and the regulatory cap gives the capped exposure $u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$ with crossing point $\bar{x} = (a_3 - a_1 a_2)/(a_3 - a_1)$. The diffusion of $x$ is $\sigma^2(x,\pi) = \pi^2\sigma^2x^2 + 2\pi c\sigma\sigma_L x(1-x) + \sigma_L^2(1-x)^2$, with $\sigma$ the asset volatility, $\sigma_L$ the liability volatility and $c$ the return-liability correlation. Buffer variance is $\zeta(x) = z'(x)^2\,\sigma^2\big(x,\pi(x)\big)$, the exposure ratio is $\omega(x) = \pi(x)\,x/(x-1)$, and $\nu_i^2 = \sigma^2/a_i^2 - 2c\sigma\sigma_L/a_i + \sigma_L^2$ for $i \in \{1,3\}$. $V$ is the value function, $y^*$ the dividend barrier, $x_L$ the recapitalisation trigger and $y_{\text{post}}$ the injection target. $W$ is the expected discounted shortfall and $N$ the discounted injection count under the optimal policy, $\theta$ is $\lambda_S$ or $K$, and $F$ is $W$ or $N$ accordingly. $\rho_L$ is the liability-net discount rate, $\mu$ the drift of $x$ along the cap and $\mu'$ its derivative in $x$, $\Lambda$ the shortfall penalty and $R$ its reference, and $\ell = (R-x)^+$ for $F = W$ and $\ell = 0$ for $F = N$. Primes are derivatives in $x$. $\kappa$ is the proportional issuance cost, $\mathscr{M}V$ the impulse operator, affine with slope $1+\kappa$ on the intervention side, and $\mu = \mu(x_L)$ in M4 and M5. In M6, $e_i$ are the discounted impulse times, $c_i$ the impulse terms and $B$ the cost bound.

### Proof of M1

*Lean:*
`Identification.vppp_at_barrier`, `Identification.fpp_at_barrier`, `Identification.barrier_slope`, `Identification.barrier_slope_pos`, `Identification.control_barrier_sign_needs_gap`, `Envelope.wode_of_envelope`, `Envelope.node_of_envelope`, `Envelope.neumann_at_ystar`.

$$\tfrac{1}{2}\sigma^2V''' + \tfrac{1}{2}(\sigma^2)'V'' + \mu V'' + \mu'V' - \rho_L V' - \Lambda' = 0, \quad V'' = 0,\ V' = 1,\ \Lambda' = 0 \ \Rightarrow\ \tfrac{1}{2}\sigma^2V''' = \rho_L - \mu'.$$

$$\tfrac{1}{2}\sigma^2F'' + \mu F' - \rho_L F + \ell = 0, \quad F'(y^*) = 0,\ \ell(y^*) = 0 \ \Rightarrow\ \tfrac{1}{2}\sigma^2F'' = \rho_L F.$$

$$V'''\frac{dy^*}{d\theta} + \partial_\theta V'' = 0, \quad \partial_\theta V'' = -F'' \ \Rightarrow\ \frac{dy^*}{d\theta} = \frac{F''}{V'''} = \frac{\rho_L F(y^*)}{\rho_L - \mu'(y^*)}.$$

Control, $\rho_L = 1$, $\mu' = 2$, $F = 1 \Rightarrow dy^*/d\theta = -1$.

□

### Proof of M2

*Lean:*
`Identification.boundary_slope`, `Identification.trigger_rises_in_lambda`, `Identification.trigger_falls_in_K`, `Identification.target_rises_in_K`, `Identification.gap_increasing_in_K`, `Identification.control_trigger_slope_needs_curvature`, `Envelope.trigger_jump`.

$$V''(x)\frac{dx}{d\theta} + \partial_\theta V'(x) = 0, \quad \partial_\theta V' = -F' \ \Rightarrow\ \frac{dx}{d\theta} = \frac{F'(x)}{V''(x)}.$$

$$\frac{dx_L}{d\lambda_S} = \frac{W'(x_L)}{V''(x_L)} > 0, \qquad \frac{dx_L}{dK} = \frac{N'(x_L)}{V''(x_L)} < 0, \qquad \frac{dy_{\text{post}}}{dK} = \frac{N'(y_{\text{post}})}{V''(y_{\text{post}})} > 0.$$

$$\frac{d(y_{\text{post}} - x_L)}{dK} > 0, \qquad N(x_L) = N(y_{\text{post}}) + 1.$$

Control, $V''(x) = 0$, $F'(x) = 0 \Rightarrow dx = 0$ and $dx = 1$ both solve the chain rule.

□

### Proof of M3

*Lean:* `Identification.det_formula`, `Identification.det_neg_of_opposite`, `Identification.det_neg`, `Identification.control_same_direction_can_fail`.

$$\det\begin{pmatrix} \frac{\rho_L W(y^*)}{\rho_L-\mu'} & \frac{\rho_L N(y^*)}{\rho_L-\mu'} \\ \frac{W'(x_L)}{V''(x_L)} & \frac{N'(x_L)}{V''(x_L)} \end{pmatrix} = \frac{\rho_L}{(\rho_L-\mu')V''(x_L)}\big(W(y^*)N'(x_L)-N(y^*)W'(x_L)\big).$$

$$a_{11},a_{12},a_{21}>0>a_{22} \;\Rightarrow\; a_{11}a_{22}<0<a_{12}a_{21} \;\Rightarrow\; a_{11}a_{22}-a_{12}a_{21}<0.$$

Control, $a_{11}=a_{12}=a_{21}=a_{22}=1 \Rightarrow \det = 0$.

□

### Proof of M4

*Lean:* `SecondOrder.psi_identity`, `SecondOrder.psi_nonneg_of_qvi`, `SecondOrder.trigger_convex`, `SecondOrder.trigger_strict_iff`, `SecondOrder.control_trigger_flat_when_psi_zero`.

$$\tfrac12\sigma^2V''+\mu V'-\rho_LV-\Lambda=0,\quad V'=1+\kappa \;\Rightarrow\; \tfrac12\sigma^2V''(x_L^+)=\rho_LV+\Lambda-(1+\kappa)\mu=\Psi(x_L).$$

$$\mu(1+\kappa)-\rho_LV-\Lambda\le 0 \;\Rightarrow\; \Psi(x_L)\ge 0 \;\Rightarrow\; V''(x_L^+)\ge 0,\qquad \sigma^2>0 \;\Rightarrow\; \big(V''(x_L^+)>0 \iff \Psi(x_L)>0\big).$$

Control, $\sigma^2=1$, $V'=1$, $\kappa=0$, $V=\Lambda=\mu=0$, $\rho_L=1 \Rightarrow \Psi=0$, $V''(x_L^+)=0$.

□

### Proof of M5

*Lean:* `SmoothFit.hasDeriv_Mv`, `SmoothFit.kink_is_convex`, `SmoothFit.no_convex_kink`, `SmoothFit.smooth_fit`, `SmoothFit.control_kink_needs_obstacle`, `Envelope.smf_permits`.

$$V\ge \mathscr{M}V,\ V(x_L)=\mathscr{M}V(x_L),\ (\mathscr{M}V)'=1+\kappa \;\Rightarrow\; V'(x_L^+)\ge 1+\kappa.$$

$$V'(x_L^+)>1+\kappa \;\Rightarrow\; \tfrac12\sigma^2q+(1+\kappa)\mu-\rho_LV-\Lambda\le 0\ \forall q,\qquad q=\frac{2(\rho_LV+\Lambda-(1+\kappa)\mu+1)}{\sigma^2} \;\Rightarrow\; 1\le 0.$$

$$\Rightarrow\; V'(x_L^+)=1+\kappa;\qquad q\le 0,\ (1+\kappa)\mu\le\rho_LV+\Lambda \;\Rightarrow\; \tfrac12\sigma^2q+(1+\kappa)\mu-\rho_LV-\Lambda\le 0.$$

Control, $v(x)=x$, $\kappa=1$, $\mathscr{M}v(x)=2x \Rightarrow v(0)=\mathscr{M}v(0)$, $v'(0^+)=1<2$.

□

### Proof of M6

*Lean:* `ImpulseCount.partial_sums_bounded`, `ImpulseCount.count_summable`, `ImpulseCount.count_le`, `ImpulseCount.impulse_sum_summable`, `ImpulseCount.control_count_needs_fixed_cost`.

$$K\sum_{i<n}e_i\le B,\ K>0 \;\Rightarrow\; \sum_{i<n}e_i\le\frac{B}{K}\ \forall n,\qquad e_i\ge 0 \;\Rightarrow\; N=\sum_i e_i\le\frac{B}{K}<\infty.$$

$$|c_i|\le Ce_i \;\Rightarrow\; \sum_i|c_i|\le CN<\infty.$$

Control, $K=0$, $e_i=1 \Rightarrow 0\le 0$ for every $n$ and $\sum_i e_i=\infty$.

□

### Proof of M7

*Lean:* `Targeting.target_free_of_start`, `Targeting.target_same_for_all_starts`, `Targeting.control_target_moves_with_convex_cost`.

$$V(z)-(1+\kappa)(z-x)-K=\big(V(z)-(1+\kappa)z\big)+\big((1+\kappa)x-K\big),$$
$$\arg\max_z\big(V(z)-(1+\kappa)(z-x)-K\big)=\arg\max_z\big(V(z)-(1+\kappa)z\big)\quad\forall x.$$

Control, $V(z)=2z$, cost $(z-x)^2 \Rightarrow \arg\max_z\big(2z-(z-x)^2\big)=x+1$.

□

### Proof of M8

*Lean:* `QVI_Part1.lambda_asymmetry_vanishes_at_one`, `QVI_Part1.neg_Lambda_concave`, `QVI_Part1.control_concavity_needs_lambda_ge_one`.

$$\Lambda(x)\big|_{\lambda_S=1}=(1-1)(R-x)^+=0.$$
$$-\Lambda(x)=(\lambda_S-1)\big(\min\{x,R\}-R\big),\qquad \lambda_S-1\ge 0,\ \ x\mapsto\min\{x,R\}\text{ concave.}$$

Control, $\lambda_S=0$, $R=0\Rightarrow -\Lambda(x)=\max\{-x,0\}$, $-\Lambda(0)=0<\frac{1}{2}=\frac{1}{2}(-\Lambda(-1))+\frac{1}{2}(-\Lambda(1))$.

□

### Proof of M9

*Lean:* `CapGeometry.zmap_inverse`, `CapGeometry.zmap_of_inverse`, `Coordinate.zmap_strictMonoOn`, `Coordinate.zmap_at_requirement`, `Coordinate.zmap_tendsto_atBot`, `Coordinate.zmap_tendsto_atTop`.

$$z(x)=\log\frac{x-1}{q},\qquad 1+qe^{z(x)}=x\ \ (x>1),\qquad z(1+qe^{w})=w.$$
$$1<x<y\ \Rightarrow\ \frac{x-1}{q}<\frac{y-1}{q}\ \Rightarrow\ z(x)<z(y),\qquad z(1+q)=\log 1=0.$$
$$\frac{x-1}{q}\to 0^+\ (x\to 1^+)\ \Rightarrow\ z\to-\infty,\qquad \frac{x-1}{q}\to\infty\ (x\to\infty)\ \Rightarrow\ z\to\infty.$$

□

### Proof of M10

*Lean:* `CapGeometry.hasDerivAt_zmap`, `CapGeometry.zvol2_eq`, `CapGeometry.xbar_sub_one`, `CapGeometry.one_lt_xbar`, `CapGeometry.solvency_binds`, `CapGeometry.liquidity_binds`, `CapGeometry.zvol2_solvency`.

$$\frac{x-1}{a_1}\le\frac{x-a_2}{a_3}\iff(a_3-a_1)x\le a_3-a_1a_2\iff x\le\bar{x},\qquad \bar{x}-1=\frac{a_1(1-a_2)}{a_3-a_1}>0.$$
$$z'(x)=\frac{1}{x-1},\qquad \zeta(x)=\omega(x)^2\sigma^2-2\,\omega(x)\,c\,\sigma\sigma_L+\sigma_L^2.$$
$$1<x\le\bar{x},\ \pi(x)x=u(x)\ \Rightarrow\ \omega(x)=\frac{1}{a_1}\ \Rightarrow\ \zeta(x)=\nu_1^2.$$

□

### Proof of M11

*Lean:* `CapGeometry.deficit_limit`, `CapGeometry.surplus_limit`, `CapGeometry.variance_ratio_is_cap_geometry`, `CapGeometry.control_surplus_limit_needs_a1_lt_a3`, `CapGeometry.control_constancy_needs_binding`.

Near $1^+$, $\bar{x} > 1$ by M10, so $1 < x < \bar{x}$ eventually and $\zeta(x) = \nu_1^2$ there. For $x \geq \bar{x}$,

$$\omega(x) = \frac{x - a_2}{a_3(x-1)} = \frac{1}{a_3} + \frac{1-a_2}{a_3}\,(x-1)^{-1} \to \frac{1}{a_3},$$

$$\zeta(x) = \omega(x)^2\sigma^2 - 2\,\omega(x)\,c\,\sigma\sigma_L + \sigma_L^2 \to \nu_3^2.$$

Controls, at $\sigma = 1$, $\sigma_L = c = 0$, $q = 1$.

$$a_1 = \tfrac{1}{2},\ a_2 = 0,\ a_3 = \tfrac{1}{4} \ \Rightarrow\ u(x) = 2(x-1),\ \zeta \equiv 4 \neq 16 = \nu_3^2.$$

$$\pi \equiv 0 \ \Rightarrow\ \zeta(2) = 0 \neq 4 = \nu_1^2.$$

□

### Proof of M12

*Lean:* `Bounds.qv_le_max`, `Bounds.zvol2_eq_qv`, `Bounds.nu2_eq_qv`, `Bounds.qv_zero`, `Bounds.deficit_bound`, `Bounds.surplus_bound`, `Bounds.control_bound_needs_long_exposure`, `Bounds.control_bound_needs_compliance`.

$$\zeta(x) = Q(\omega), \qquad Q(\omega) = \omega^2\sigma^2 - 2\omega c\sigma\sigma_L + \sigma_L^2, \qquad \omega = \frac{\pi(x)\,x}{x-1},$$

$$Q(0) = \sigma_L^2, \qquad Q(1/a_i) = \nu_i^2.$$

$$\omega = tM,\ t \in [0,1] \ \Rightarrow\ Q(\omega) = (1-t)Q(0) + tQ(M) + \sigma^2M^2(t^2 - t) \leq \max\{Q(0), Q(M)\}.$$

$$0 \leq \pi,\ \ \pi x \leq u(x) \leq \frac{x-1}{a_1} \ \Rightarrow\ 0 \leq \omega \leq \frac{1}{a_1} \ \Rightarrow\ \zeta(x) \leq \max\{\sigma_L^2, \nu_1^2\}.$$

$$\pi x \leq \frac{x - a_2}{a_3} \ \Rightarrow\ \omega \leq w(x) = \frac{1}{a_3} + \frac{1-a_2}{a_3(x-1)} \to \frac{1}{a_3}, \qquad Q(w(x)) \to \nu_3^2.$$

Controls, $(\sigma, \sigma_L, c) = (1,1,1)$, $\omega = -1 \Rightarrow Q = 4 > 1 = \max\{Q(0), Q(1)\}$; $(\sigma, \sigma_L, c) = (1,0,0)$, $\omega = 2 > 1 \Rightarrow Q = 4 > 1$.

□

### Proof of M13

*Lean:* `Comparison.nu1_baseline`, `Comparison.nu3_baseline`, `Comparison.ratio_baseline`.

$$\nu_1^2 = \frac{0.08^2}{0.045^2} - \frac{2(0.20)(0.08)(0.03)}{0.045} + 0.03^2 = \frac{256}{81} - \frac{16}{750} + \frac{9}{10000} = \frac{2543449}{810000}.$$

$$\nu_3^2 = \frac{0.08^2}{0.30^2} - \frac{2(0.20)(0.08)(0.03)}{0.30} + 0.03^2 = \frac{16}{225} - \frac{4}{1250} + \frac{9}{10000} = \frac{6193}{90000}.$$

$$\frac{\nu_1^2}{\nu_3^2} = \frac{2543449}{810000} \cdot \frac{90000}{6193} = \frac{2543449}{55737}.$$

□

### Proof of M14

*Lean:* `Comparison.nu2_hedged`, `Comparison.hedged_is_cushion_vol`, `Comparison.hedged_baseline`, `Comparison.control_liability_shock_moves_nu`.

$$\sigma_L = 0 \;\Rightarrow\; \nu^2 = \frac{\sigma^2}{a^2} = \Big(\frac{1}{a}\Big)^2 \sigma^2.$$

$$\frac{1}{a}(W-F) \le bW,\; W-F>0 \;\Rightarrow\; \frac{E}{C} = \frac{1}{a}, \quad \Big(\frac{E}{C}\Big)^2 \sigma^2 = \Big(\frac{1}{a}\Big)^2 \sigma^2 = \nu^2\big|_{\sigma_L=0}.$$

$$\Big(\frac{0.08}{0.045}\Big)^2 = \frac{256}{81}, \qquad \Big(\frac{0.08}{0.30}\Big)^2 = \frac{16}{225}.$$

Control, $(\sigma, \sigma_L, c, a) = (1,1,0,1) \Rightarrow \nu^2 = 1-0+1 = 2 \neq 1$.

□

### Proof of M15

*Lean:* `TimeVarying.deficit_limit_at_date`, `TimeVarying.same_variance_any_binding_policy`, `TimeVarying.surplus_limit_at_date`, `TimeVarying.nu2_of_variance`, `TimeVarying.control_date_identity_needs_binding`.

M10 and M11 are statements at one value of the asset volatility, so at each date $t$ they hold with $\sigma = \sigma_t$,

$$\zeta_t(x) = \omega_t(x)^2 \sigma_t^2 - 2\,\omega_t(x)\, c\, \sigma_t \sigma_L + \sigma_L^2, \qquad \omega_t(x) = \frac{\pi_t(x)\, x}{x-1}.$$

Binding on the solvency branch gives $\omega_t = 1/a_1$, so $\zeta_t = \nu_1^2$ at $\sigma_t$, and two exposures that bind there have the same $\omega_t$ and so the same $\zeta_t$. Deep in surplus $\omega_t \to 1/a_3$ as in M11. Writing $\sigma_t^2$ for the asset variance gives the form in it. Control. With $\pi \equiv 0$, $\omega = 0$ and $\zeta = \sigma_L^2 = 1$, against $\nu_1^2 = 1/(1/2)^2 + 1 = 5$.

□

## B Evidence and verification

Every row of the model table is a theorem of Lean 4 with Mathlib. The Lean column names the theorems, and the proof of each row in Appendix A names the same ones. A theorem cited as `Namespace.name` is declared in the file `theory/lean/Namespace.lean`, and `theory/lean/ORIGIN.md` pins each file by its sha256. The development is built on the toolchain `v4.32.0-rc1` against Mathlib commit `360da6f`. Every literature row rests on a record under `theory/lit/`, whose suite checks each quotation on its page of the pinned source and builds and audits the record's Lean file. The script `theory/verify.sh` runs the suites below on every build, and `theory/VERIFICATION.md` records what they printed.

1. *The Lean build and axiom audit.* Every file under `theory/lean/` is a build root and builds. No source contains `sorry`. Every declared theorem is audited with `#print axioms`, and none depends on an axiom outside `propext`, `Classical.choice` and `Quot.sound`. Each file matches its sha256 pin in `theory/lean/ORIGIN.md`.

2. *The skeleton checker.* Rules MS-1 to MS-10 of `theory/checks/verify_manuscript.py`, each also run on a well-formed sample it must accept and a faulty sample it must reject.

3. *The literature records.* The suite of every record a literature row cites.

**What Lean does not carry.** Buffer variance $\zeta$ is defined as $z'(x)^2\sigma^2(x,\pi(x))$, and that this is the diffusion coefficient of $z(X_t)$ is Itô's formula, which is not proved (A3). That the optimal policy binds the cap near both ends of the state space is a hypothesis of M11, which K2 and C2 state as their condition, which the solve of the Illustrations paragraph supports at one parameter point.

M1 to M3 take as hypotheses, in raw form, the inputs of A6. That $V$ is the value function is the verification theorem, not proved. Differentiability of $V$ and of the thresholds in $\theta$ is the regularity lemma and the implicit function theorem. The envelope identities $\partial_{\lambda_S} V = -W$ and $\partial_K V = -N$ are Danskin's theorem. The interior equation enters differentiated in $x$ and in $\theta$. The signs $W(y^*) > 0$, $N(y^*) > 0$, $N' < 0$ on the inaction region and $W'(x_L) > 0$ come from hitting-time arguments, and $W'(x_L) > 0$ holds exactly when $\rho_L W(y_{\text{post}}) < R - x_L$. $V''(x_L) > 0$ follows from M4 given $\Psi(x_L) > 0$, the strict form of the inequality the quasi-variational inequality gives on the intervention side. $V''(y_{\text{post}}) < 0$ is a hypothesis, since the derivation offered for it fails (Appendix E). $\rho_L > \mu'(y^*)$ is a parameter condition. K1 and C1 state these as their hypotheses.

M5 takes the viscosity supersolution property of $V$ at $x_L$ as a hypothesis, in the form that a kink admits lower test functions of every curvature. M6 takes the cost bound as a hypothesis, that each impulse costs at least $K$ and the strategy's value is finite, with $e_i = e^{-\rho_L \tau_i}$ the discounted impulse times and $c_i$ the impulse terms of the verification argument. Interior regularity of $V$, $C^2$ on the inaction region and $C^3$ away from $R$ and $\bar{x}$, is assumed wherever a row differentiates $V$.

**Illustrations.** No row rests on SymPy or on numerical sampling. That the hypotheses of M1 to M4 hold at some parameter point is shown by a numerical solve, `theory/solve/solve_qvi.py`, whose output is `theory/solve/results_solve.md`. The solve takes the baseline parameters of Bayraktar et al. [2026] with the riskless rate $r = 0.025$ in place of their 0.02, $\lambda_S = 1.2$ and $K = 0.05$, and solves the discrete quasi-variational inequality by policy iteration on grids of 601 and 1201 nodes, to a residual below $10^{-12}$. On the finer grid the dividend barrier is $y^* = 1.741$, the recapitalisation trigger $x_L = 1.026$ and the injection target $y_{\text{post}} = 1.553$, each within 0.002 of its value on the coarser grid. Every hypothesis of M1, M2 and M4 holds there, among them $W(y^*) = 0.079 > 0$, $W'(x_L) = 1.84 > 0$, $N' < 0$ on the inaction region, $\Psi(x_L) = 0.027 > 0$, $V''(y_{\text{post}}) < 0$ and $\mu'(y^*) = 0.045 < \rho_L = 0.09$. The envelope identities hold to within 0.7 percent, and the responses of the thresholds by finite differences have the signs of M1 and M2, with a determinant of $-0.30$ against $-0.33$ from the formula of M3. The risky exposure is at the cap at both thresholds and on 85 percent of the inaction region, and below it on an interval around the crossing point. Over 56 values of $(\lambda_S, K)$, with $\lambda_S$ from 1.02 to 2 and $K$ from 0.01 to 0.2, the trigger is interior at 42 and every hypothesis holds there. At the other 14, where the fixed issuance cost is large against the asymmetry parameter, the bank issues only at the distress boundary, so the trigger cannot move and the identification of K1 does not apply.

**Without borrowing.** The solve above holds the risky exposure at $\pi$ between 3.2 and 3.3 over most of the inaction region, the leveraged case of C3. The same solve with the exposure limited to $\pi \leq 1$, the main banking case of Bayraktar et al. [2026] (L2), is `theory/solve/run_noborrow.py`, whose output is `theory/solve/results_noborrow.md`. At $(\lambda_S, K) = (1.2, 0.05)$ every hypothesis of M1, M2 and M4 holds on both grids, with $y^* = 1.251$, $x_L = 1.006$ and $y_{\text{post}} = 1.218$ on the finer, and the responses of the thresholds by finite differences have the signs of M1 and M2. Over the same 56 values of $(\lambda_S, K)$ the trigger is interior at 27, where every hypothesis holds, and at the distress boundary at the other 29. The solvency branch binds only up to $x = 1/(1-a_1) \approx 1.047$, and above it the limit on the exposure is $\pi \leq 1$ rather than the liquidity branch, so the exposure is on the solvency branch next to the trigger, at $\pi = 1$ next to the dividend barrier, and below both on an interval between them. Buffer variance equals the deficit limit $v_1^2 = 3.14$ on $[x_L, 1.047]$ and stays within the bound of M12 on the whole inaction region. At the dividend barrier it is 0.155 without borrowing and 0.364 with it, against a surplus limit of 0.0688, since the state does not rise above the dividend barrier.

**Comparative statics.** The comparative statics are in Section 3, and Table 6 lists the script and the saved output behind each of its figures, tables and numbers.

# C Terminology and measurement map

The map names every object of the model once, and the manuscript uses these names and no others. It is `theory/MEASUREMENT_MAP.tex`, whose tables this appendix includes. The names are those of the map's Names

table, agreed with the author on 10, 13 and 19 October 2026, and four names the author accepted on 23 October 2026, the three parameters of the cap geometry taken one by one and the value function. The objects fall into three tables. Inputs are what the model takes as given, internal objects are built from the inputs inside the model, and outputs are what the rows predict. The column Directly observed says Yes when a record of the world gives the object without any computation, which is when its last column begins with Recorded, and No otherwise. Every symbol in the table of assumptions has exactly one row here. Where a row names the panel, it is the FDIC panel of US banks, 2015 to 2024, that the author's earlier empirical work read, and the figures are from that reading.

The map carries the reassessment of the question. The asymmetry parameter and the fixed issuance cost are never observed. The two thresholds that identify them are recorded as the state at a dividend and at an issue, and on the panel read neither behaves as a threshold. The limits of buffer variance are computed from assumed volatilities and apply where the cap binds, and on that panel it does not. What reaches the data is the bound of M12 and the two recorded thresholds as events, so K1 and K2 are results within the model, and the question is one of identification, not of measurement.

## C.1 Inputs

Table 14: Inputs of the model.

| Symbol | Name | Type | Represents | Directly observed | Obtained how |
|---|---|---|---|---|---|
| $x$ | **Assets-to-liabilities ratio** | A real number above one. | Total assets over total liabilities, the state of the model (A1). | Yes | Recorded as total assets over total liabilities in the Call Report, `ASSET` over `LIAB`. |
| $\pi$ | **Risky exposure** | A function of the state with values between zero and the cap over the state. | The share of assets in the risky asset (A1, A2). No row derives it; M11 takes it at the cap and M12 bounds every compliant choice. | Yes | Recorded as risk-weighted assets over total assets, `RWAJT` over `ASSET`, under risk weights of zero on the riskless and one on the risky asset. On the panel its median is 0.69 against a cap of 2.38 at the median state, so the cap does not bind there. |
| $D$ | **Dividends** | A non-decreasing process, zero before the first date. | Cumulative payments to shareholders per unit of liabilities, the singular control of the bank's problem. Under the optimal policy it rises only when the state is at the dividend barrier (A6). | Yes | Recorded as cash dividends declared, `EQCDIV`, over liabilities. |
| $I_n$ | **Recapitalisation** | A positive real number for each issue. | Capital raised in the $n$-th issue per unit of liabilities, the impulse control, at the issue time $\tau_n$. Under the optimal policy it is the injection target less the recapitalisation trigger (A6). | Yes | Recorded as net stock issuance (`EQCSXQ`) and capital from the parent holding company (`EQCBHCTR`), over liabilities. |
| $r$ | **Riskless rate** | A positive real number. | The return on the riskless asset, in the drift of the state (A1). | No | Assumed at 0.025 in the paper's illustrations and at 0.02, the baseline of Bayraktar et al. [2026], in its comparative statics. |
| $\mu_s$ | **Expected asset return** | A real number. | The drift of the risky asset, in the drift of the state (A1). | No | Assumed at the baseline of Bayraktar et al. [2026], 0.04. |
| $\mu_L$ | **Liability drift** | A real number. | The drift of liabilities, in the drift of the state and in the liability-net discount rate, which is the discount rate less the liability drift (A1, A4). | No | Assumed at the baseline of Bayraktar et al. [2026], 0.03. |

| Symbol | Name | Type | Represents | Directly observed | Obtained how |
|---|---|---|---|---|---|
| $\gamma$ | **Net income on liabilities** | A real number. | Income earned on liabilities net of their cost, per unit of liabilities, in the drift of the state (A1). | No | Assumed at the baseline of Bayraktar et al. [2026], 0.02. They read the same term as the exogenous growth of deposits, which adds to assets. |
| $\sigma$ | **Asset volatility** | A positive real number. | The volatility of the risky asset, and the letter of the diffusion of the state (A1, A3). | No | Assumed at the baseline of Bayraktar et al. [2026], 0.08. The volatility of the book ratio is computed on the panel and is 4.1 times smaller than the model's at the recorded exposure, so it is not this object. |
| $\sigma_L$ | **Liability volatility** | A positive real number. | The volatility of the shock to liabilities (A1). | No | Assumed at the baseline of Bayraktar et al. [2026], 0.03. |
| $c$ | **Return-liability correlation** | A real number of absolute value below one. | The correlation between the risky return and the liability shock (A1). | No | Assumed at the baseline of Bayraktar et al. [2026], 0.20. |
| $q$ | **Required capital** | A positive real number. | The capital regulation requires, as a ratio to liabilities, the zero of the capital buffer (A1). | No | Assumed. A floor on the leverage ratio gives a candidate, 3/97 for the 3% Tier 1 leverage ratio when Tier 1 equals book equity, and the paper does not fix it. Buffer variance does not depend on it. |
| $a_1$ | **Solvency floor** | A real number between zero and the liquidity haircut. | The floor on equity over risky assets, the slope of the solvency branch of the regulatory cap (A2). | Yes | Recorded in the Basel standard as the CET1 minimum of 4.5% of risk-weighted assets, the baseline of Bayraktar et al. [2026]; which tier it stands for is not fixed. |
| $a_2$ | **Run-off rate** | A real number below one. | Net 30-day outflows as a share of liabilities, the intercept of the liquidity branch (A2). | No | Assumed at 0.05, the standard's run-off factor for stable retail deposits. The standard sets a factor per class of liability, so one rate for all liabilities is the model's reduction. |
| $a_3$ | **Liquidity haircut** | A real number between the solvency floor and one. | The haircut on the risky asset in high-quality liquid assets, the slope of the liquidity branch (A2). | No | Assumed at 0.30, the baseline of Bayraktar et al. [2026], which is none of the standard's haircuts. |
| $\rho_L$ | **Liability-net discount rate** | A positive real number. | The discount rate of the problem written per unit of liabilities (A4). | No | Assumed from the shareholders' discount rate, 0.12 at the baseline of Bayraktar et al. [2026]. |
| $\lambda_S$ | **Asymmetry parameter** | A real number at least one. | How much more the objective weighs a shortfall below the reference than a surplus above it (A4). At one the shortfall penalty vanishes and the model has issuance costs alone (M8). | No | Never observed. It enters the rows through the shortfall penalty only. M3 identifies it locally with the fixed issuance cost from the dividend barrier and the recapitalisation trigger, a statement within the model, since on the panel neither threshold is measured as one (the rows of the dividend barrier and the recapitalisation trigger below). |

| Symbol | Name | Type | Represents | Directly observed | Obtained how |
|---|---|---|---|---|---|
| $K$ | **Fixed issuance cost** | A positive real number. | The cost of an issue that does not scale with its size, per unit of liabilities (A5). | No | Never observed. Its counterpart is the part of an issue's cost that does not scale with the amount raised, which Peura and Keppo [2006] set from implied issue cost and frequency rather than record (L8). M3 identifies it locally with the asymmetry parameter from the two thresholds. |
| $\kappa$ | **Proportional issuance cost** | A non-negative real number. | The cost of an issue per unit raised (A5). | No | Assumed at the baseline of Bayraktar et al. [2026], 0.01 for dilution and 0.02 per unit issued. |

## C.2 Internal objects

Table 15: Internal objects of the model.

| Symbol | Name | Type | Represents | Directly observed | Obtained how |
|---|---|---|---|---|---|
| $z$ | **Capital buffer** | A real number, $z = \log((x-1)/q)$. | The log of equity over required capital, zero where equity equals required capital and falling without limit towards the distress boundary (A1, M9). | No | Computed from the assets-to-liabilities ratio and required capital by its formula, whose inverse is `CapGeometry.zmap_inverse` and which is zero where equity equals required capital, `Coordinate.zmap_at_requirement`. The author's earlier empirical work records its discrete-time form as Tier 1 capital over 8.5% of risk-weighted assets. |
| $R$ | **Reference** | A real number, $R = 1+q$. | The state at which equity equals required capital, below which the shortfall penalty applies (A4). The capital buffer is zero there. | No | Computed from required capital by its formula. The capital buffer vanishes at it, `Coordinate.zmap_at_requirement`; the penalty is zero at or above it, `QVI_Part1.shortfall_zero_at_or_above_reference`, and positive below it for an asymmetry parameter above one, `QVI_Part1.shortfall_pos_below_reference`. Its counterpart is the required capital of the solvency floor's row, the level at which prompt corrective action's restrictions begin. |
| $u$ | **Regulatory cap** | A function of the state, $u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$. | The largest risky exposure regulation allows at each state, the smaller of its two branches (A2). | No | Computed from the assets-to-liabilities ratio and the cap geometry by its formula. The solvency branch is the smaller up to the crossing point, `CapGeometry.solvency_binds`, and the liquidity branch beyond it, `CapGeometry.liquidity_binds`. |

| Symbol | Name | Type | Represents | Directly observed | Obtained how |
|---|---|---|---|---|---|
| $\bar{x}$ | **Crossing point** | A real number above one, $\bar{x}-1=a_1(1-a_2)/(a_3-a_1)$. | The state at which the two branches of the regulatory cap are equal (M10). | No | Computed from the cap geometry by its formula, `CapGeometry.xbar_sub_one`. |
| $\mu$ | **Drift of the state** | A function of the state and the risky exposure, $\mu(x,\pi)=x((1-\pi)r+\pi\mu_s-\mu_L)+\gamma$. | The expected change per unit of time of the assets-to-liabilities ratio between issues and dividends (A1). Along the cap it is the drift whose derivative at the dividend barrier enters M1. | No | Computed from the riskless rate, the expected asset return, the liability drift and the net income on liabilities. |
| $\Lambda$ | **Shortfall penalty** | A function of the state, $\Lambda(x)=(\lambda_S-1)(R-x)^+$. | The term of the objective that charges a shortfall below the reference (A4). Zero when the asymmetry parameter is one, `QVI_Part1.lambda_asymmetry_vanishes_at_one`. | No | Never observed. It is a term of the objective, and whose cost it stands for is a reading, the shortfall-penalty paragraph of the manuscript's primitives appendix. |
| $V$ | **Value function** | A function of the state, twice continuously differentiable on the inaction region (A6). | The value of the bank under its best policy, per unit of liabilities. | No | Never observed. Its counterpart would be the market value of equity per unit of liabilities, which the panel does not carry; Dai et al. [2019] fit their parameters to such values (L21). |
| $\mathscr{M}V$ | **Impulse operator** | A function of the state, the supremum over targets of the value after an issue net of both issuance costs. | The value of issuing at once to the best target (A6, M5, M7). | No | A device. It exists only inside the quasi-variational inequality and the proofs of M5 and M7. |
| $W$ | **Expected discounted shortfall** | A function of the state, $W=-\partial_{\lambda_S}V$. | The expected discounted shortfall below the reference under the optimal policy (A6). | No | A device. It enters M1 to M3 through the envelope identity only. |
| $N$ | **Discounted injection count** | A function of the state, $N=-\partial_K V$. | The expected discounted number of issues under the optimal policy (A6). | No | A device. It enters M1 to M3 through the envelope identity only; its sample counterpart, a discounted count of recorded issues, is used by no row. |

## C.3 Outputs

Table 16: Outputs of the model.

| Symbol | Name | Type | Represents | Directly observed | Obtained how |
|---|---|---|---|---|---|
| $y^*$ | **Dividend barrier** | A real number above one. | The state at which dividends are paid (A6). M1 gives its response to the asymmetry parameter and the fixed issuance cost. | Yes | Recorded as the state at which a dividend is paid, from dividend quarters (`EQCDIV`) and the recorded state. On the panel, 56% of bank-quarters below the solved barrier pay a dividend and payment falls as the state rises, so no barrier is measured there, and under the conservation buffer the recorded payouts reflect a rule as well as the barrier. |
| $x_L$ | **Recapitalisation trigger** | A real number above one and below the dividend barrier. | The state at which capital is raised (A6). M2 gives its response to the two parameters. | Yes | Recorded as the state just before a recapitalisation, from net stock issuance (`EQCSXQ`) and capital from the parent holding company (`EQCBHCTR`). On the panel, issues start at a mean state of 1.11 against a solved trigger of 1.03. |
| $y_{\text{post}}$ | **Injection target** | A real number between the trigger and the barrier. | The state an issue reaches (A6). M7 says it does not depend on where the issue starts. | Yes | Recorded as the state just after a recapitalisation. On the panel the post-issue state rises with the pre-issue state, against the single target of M7. |
| $\zeta$ | **Buffer variance** | A function of the state, $z'(x)^2$ times the diffusion $\sigma^2(x,\pi(x))$. | The variance per unit of time of the capital buffer (A3). | No | Computed from the capital buffer, the asset volatility, the liability volatility, the return-liability correlation and the risky exposure by its formula, `CapGeometry.zvol2_eq`. Not computed, since a level needs the asset volatility, which is assumed. |
| $v_1^2$ | **Deficit limit** | A real number, $v_1^2 = \sigma^2/a_1^2 - 2c\sigma\sigma_L/a_1 + \sigma_L^2$. | The limit of buffer variance at the distress boundary where the exposure binds the solvency branch (M10, M11). | No | Computed from the asset volatility, the liability volatility, the return-liability correlation and the solvency floor by its formula, `CapGeometry.deficit_limit`. It applies where the cap binds, which on the panel it does not (the row of the risky exposure). |
| $v_3^2$ | **Surplus limit** | A real number, $v_3^2 = \sigma^2/a_3^2 - 2c\sigma\sigma_L/a_3 + \sigma_L^2$. | The limit of buffer variance deep in surplus where the exposure binds the liquidity branch (M11), which needs borrowing (C3). | No | Computed from the asset volatility, the liability volatility, the return-liability correlation and the liquidity haircut by its formula, `CapGeometry.surplus_limit`. It applies in the leveraged case only (C3). |

# D Primitives and scope

**The balance sheet.** The bank holds a risky asset with volatility $\sigma$ and a riskless asset, and owes liabilities with their own shock of volatility $\sigma_L$ and correlation $c$ with the risky return (A1). The riskless asset does not track the liabilities, unlike the reserve asset of Black and Perold [1992], so the liability shock enters buffer variance unhedged. Rows M9 to M12 depend on it.

**The regulatory cap.** The capped exposure is $u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$ with $0 < a_1 < a_3$ and $a_2 < 1$ (A2), the cap of Bayraktar et al. [2026]. Under these restrictions the solvency branch binds near the distress boundary and the liquidity branch deep in surplus (M10). With $a_1 \geq a_3$ the surplus limit of M11 fails, by its control. With no borrowing, $u(x) \leq x$, the liquidity branch lies above the ceiling wherever $a_2 \leq (1-a_3)x$ (`BCVW.ceiling_tighter_than_lcr`, in the record of L2), so the surplus limit of M11 is a statement about the leveraged case. Rows M10, M11, M12, K2, C2 and C3 depend on it.

**The fixed issuance cost.** $K$ is charged per unit of liabilities, so the dollar cost of an issue is $K$ times liabilities (A5). This keeps the reduction of the bank's problem to the single state $x$, which a fixed issuance cost in dollars would break (L4). M1 to M3, K1 and C1 depend on it.

**The shortfall penalty.** The objective charges $\Lambda(x) = (\lambda_S - 1)(R-x)^+$ below the reference $R$ (A4). The reference is $R = 1+q$, the state at which equity equals required capital and the capital buffer is zero (`Coordinate.zmap_at_requirement`), so the penalty is charged exactly in deficit, zero at or above required capital (`QVI_Part1.shortfall_zero_at_or_above_reference`) and positive below it once $\lambda_S > 1$ (`QVI_Part1.shortfall_pos_below_reference`). This is the author's choice of 23 October 2026, and it makes the hypothesis $\Lambda'(y^*) = 0$ of M1 the statement that the dividend barrier lies above the reference. The model does not say whose cost the penalty is. The reading adopted is that it stands for the supervisor's restrictions below a capital threshold, which no row derives. What K1 identifies is the asymmetry parameter of that penalty, under this reading. Peura and Keppo [2006] name which constituency binds the bank's capital as the issue their omissions reduce to (L19), and Milne and Whalley [2001] name the supervisor's intervention and the market's response together (L16). The thresholds identify the asymmetry parameter under any of these readings and do not choose among them. K1, C1 and C6 depend on this paragraph.

**The objective.** The papers read write a bank's objective in three ways. Peura and Keppo [2006], Milne and Whalley [2001] and Bayraktar et al. [2026] maximise the shareholders' value, linear in dividends and in what the shareholders pay for issues, so that any aversion to risk is induced by the frictions (L68). Milne and Whalley [2001] name the alternative, a utility concave in the bank's cash flows that is justified by a divergence of interest between managers and shareholders, and report that in portfolio models with such a utility a capital requirement can shift the bank towards riskier assets (L66, L67). The preference models put the curvature at a reference point instead, loss aversion over gains and losses in Barberis et al. [2001] and Li et al. [2023], and a floor under the portfolio in Black and Perold [1992] (L36, L14, L5). The bank papers that charge a shortfall write the charge as an amount that is the same at every shortfall below a buffer requirement, or as a smooth function of net worth that also rewards a surplus (L53, L51, L56).

The paper keeps the shareholders' value and adds one kink. Its objective is linear in dividends and in issue costs, as in the models with issuance costs, and charges $(\lambda_S - 1)(R-x)^+$ per unit of time below required capital (A4). Linearity in dividends keeps the dividend barrier a barrier, since it gives the term $1 - V'$ of equation (1), where a utility concave in the dividend rate would pay at every state the rate whose marginal utility equals $V'$. At $\lambda_S = 1$ the objective is that of the models with issuance costs, so the paper's model nests them and the asymmetry parameter measures the departure from them, and for every $\lambda_S \geq 1$ the running payoff is concave (M8). Charged per unit of liabilities, the penalty keeps the problem to one state (L4). The kink puts the asymmetry parameter in the payoff term alone, away from the cap term, which K2 uses (Section 4.3), and the penalty is zero above the reference, which M1 uses at the dividend barrier. A utility concave in cash flows would add a curvature parameter that moves the dividend barrier as the asymmetry parameter does, and the thresholds would then have to separate the two. The paper does not take that route, and whether the aversion to risk of a concave utility and that induced by the kink lead the bank to different policies is not asked.

**The policy.** That the optimal policy is a dividend barrier, a recapitalisation trigger and an injection target, and that $V$ is the value function, is the verification theorem, taken as given and not proved here (A6). M1 to M5 depend on it.

# E Refuted conjectures

**R1, strict concavity at the injection target.** An earlier draft of the proofs derived $V''(y_{\text{post}}) < 0$ from the sign change of $\varphi' = V' - (1+\kappa)$ at $y_{\text{post}}$, positive below and negative above, and from $\varphi \in C^3$, arguing that $\varphi''(y_{\text{post}}) = 0$ would give a zero of even order. A zero of odd order can still have $\varphi'' = 0$. With $\varphi(y) = -y^4/4$ and $y_{\text{post}} = 0$, $\varphi'(y) = -y^3$ is positive for $y < 0$ and negative for $y > 0$, while $\varphi''(0) = 0$ (`SecondOrder.refuted_target_strict`). The weak inequality survives and strictness does not, so $V''(y_{\text{post}}) < 0$ enters M2 as a hypothesis (Appendix B). C1 does not use it.

# F Responses to the referee reports

Not applicable. The paper has not been through review.

# G Coverage of the referee comments

Not applicable, for the same reason.